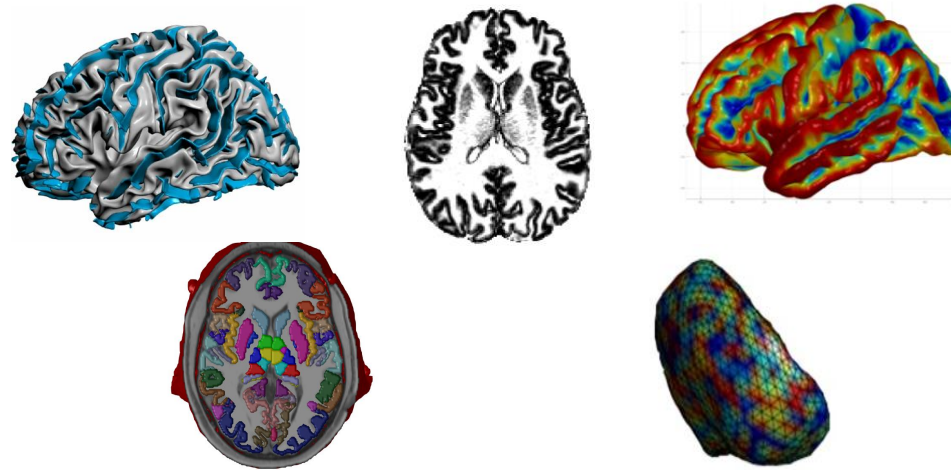




MRI Processing Theory: Structural Morphometry

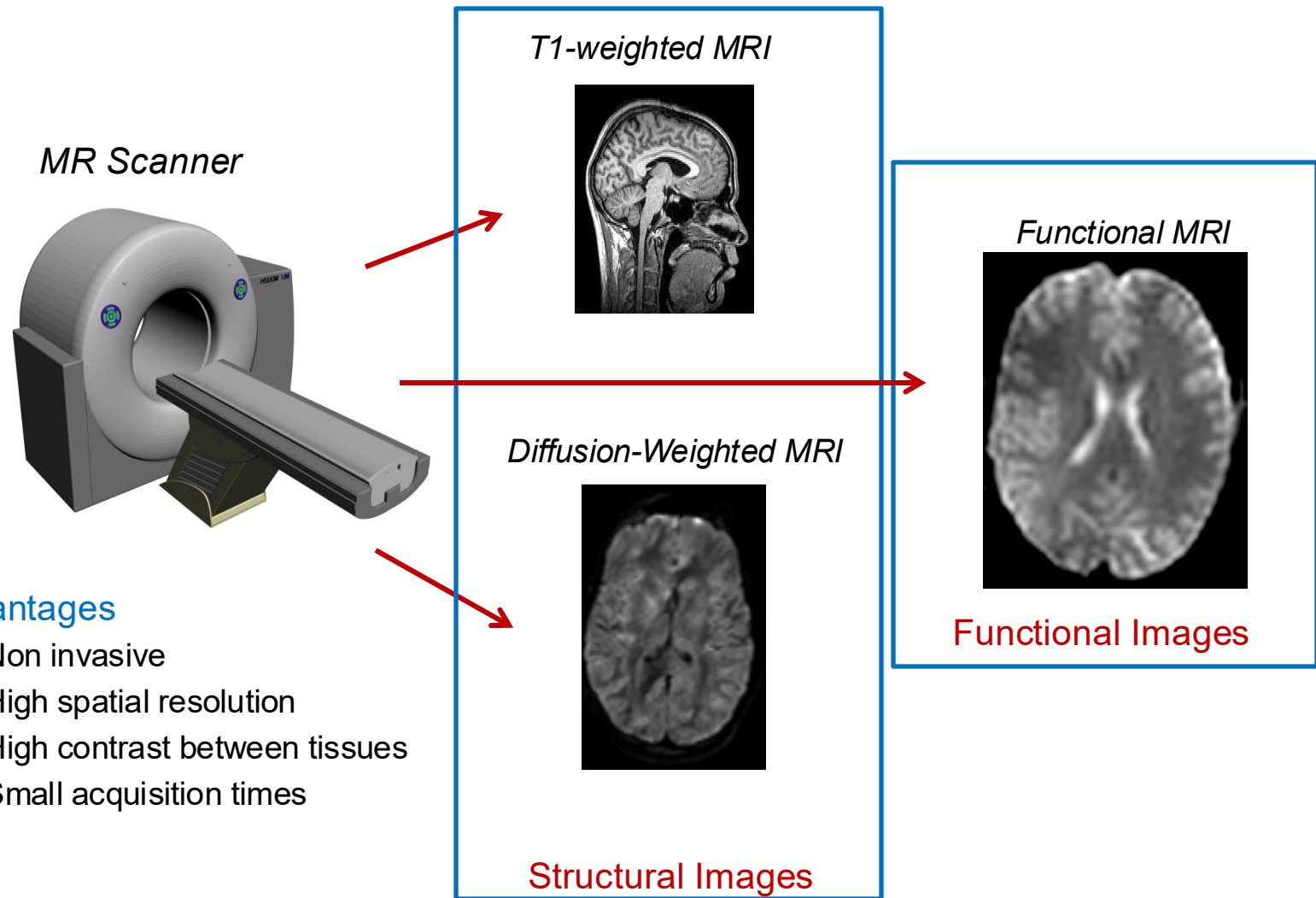
Yasser Alemán-Gómez

Radiology department. CHUV, Switzerland
October 1st, 2026

Outline

- Introduction to MRI Processing
- Region of Interest (ROI) Analysis
- Voxel-Based Morphometry (VBM)
- Surface Based Morphometry (SBM)

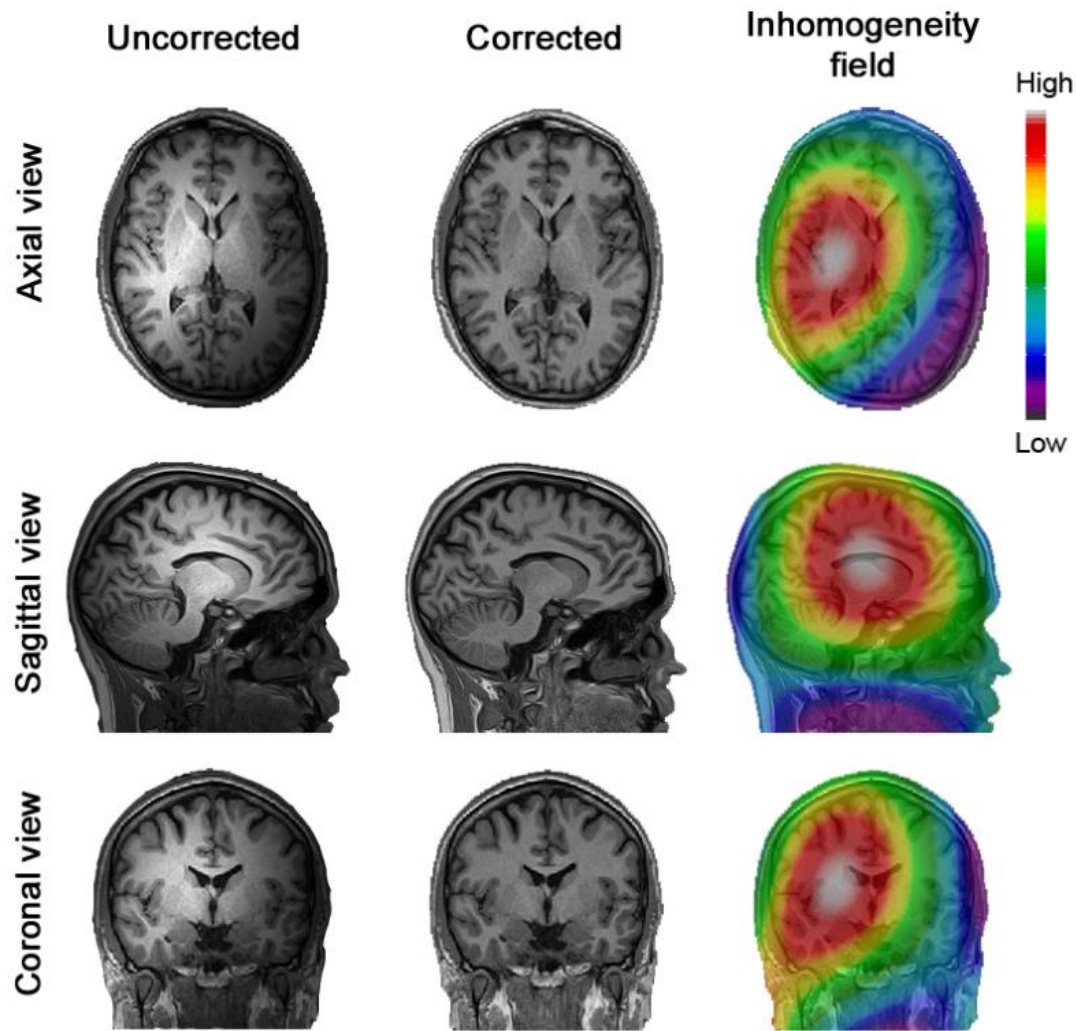
Introduction: MRI processing



■ Advantages

- ❑ Non invasive
- ❑ High spatial resolution
- ❑ High contrast between tissues
- ❑ Small acquisition times

Bias field correction



ANTS
(*N4BiasFieldCorrection*, C++)

Region of Interest (ROI) Analysis

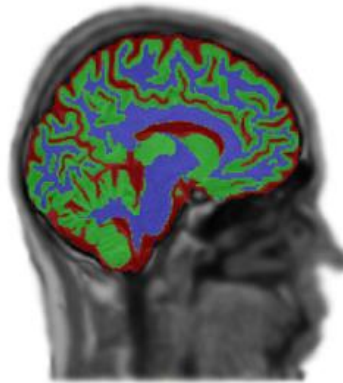
T1-Weighted image



Tissue
Segmentation



Brain tissues



Available methods

FSL

(fast, C++)

ANTs

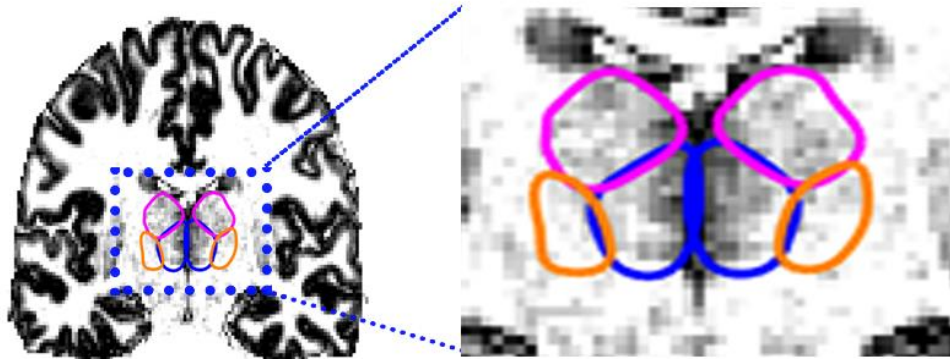
(Atropos, C++)

SPM

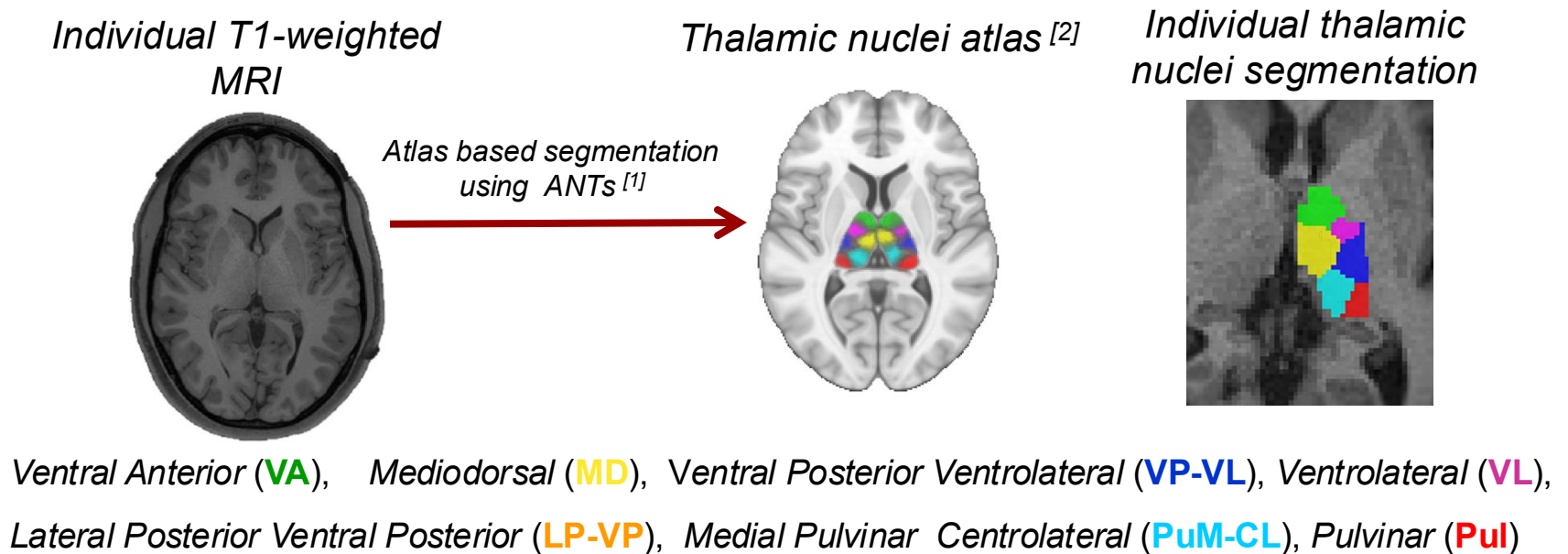
(spm_segment, Matlab)

NiSeg

(partial_volume_estimation, Python)



Region of Interest (ROI) Analysis



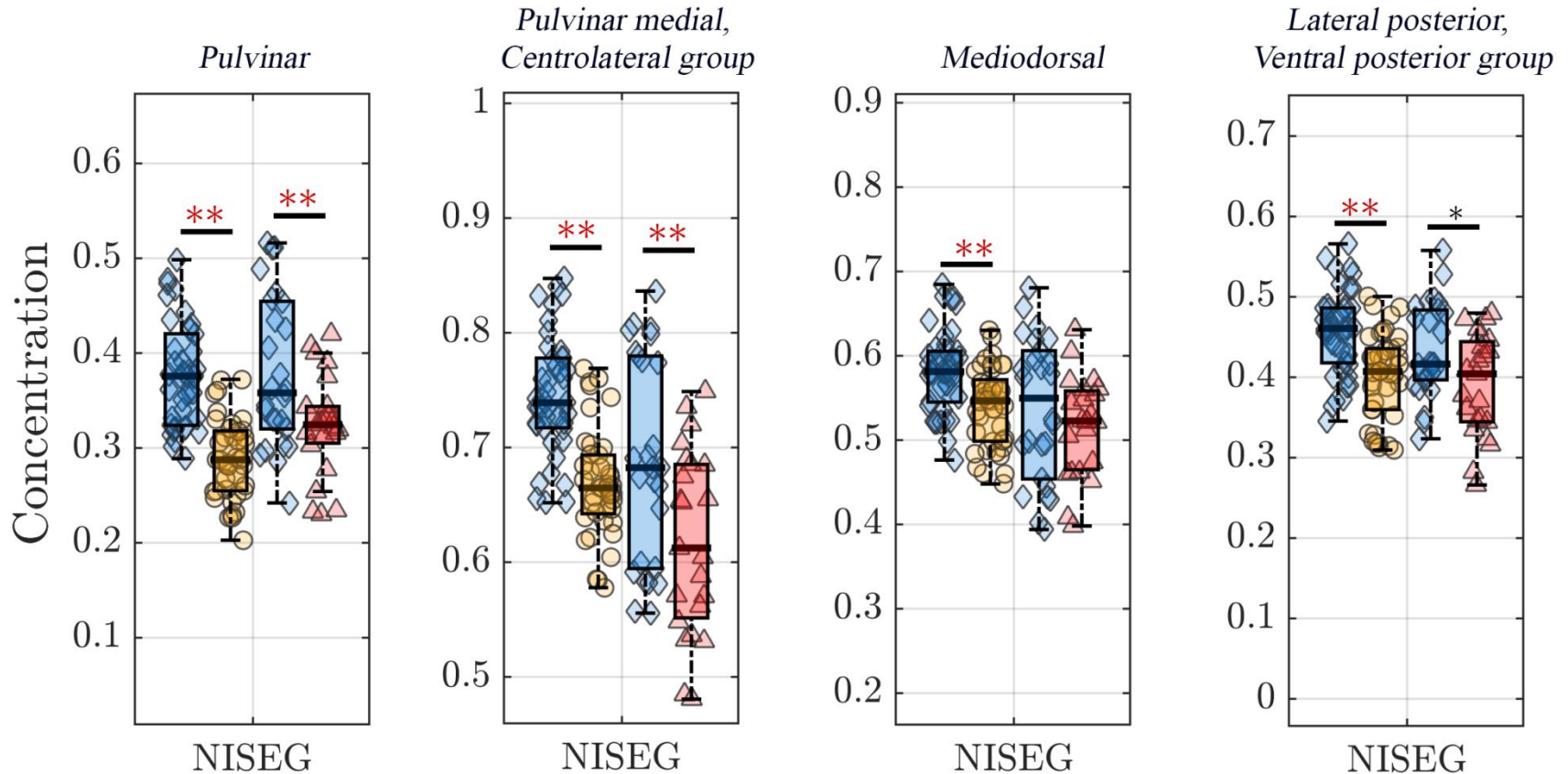
■ Advantages

- It works in subject space

■ Drawbacks

- Previously defined ROI

Group differences in gray matter for the thalamic nuclei



◆ Healthy controls

● Patients with early psychosis

▲ Patients with schizophrenia

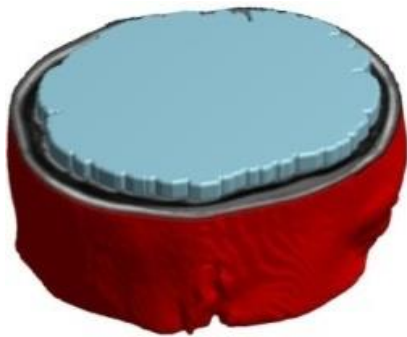
* Significant differences between groups before FDR correction

** Differences that remained significant after the FDR correction

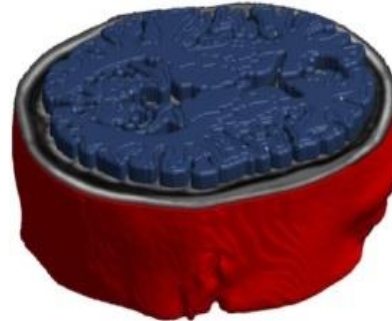
Alemán-Gómez Y, Human Brain Mapping, 2020

Region of Interest (ROI) Analysis

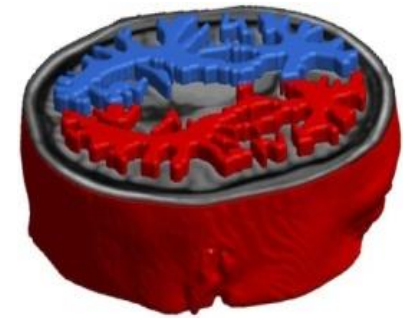
*Intracranial Volume
(WM+GM+CSF)*



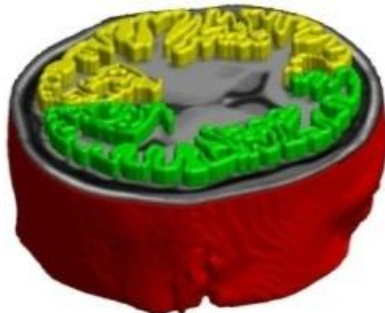
*Brain Volume
(WM+GM)*



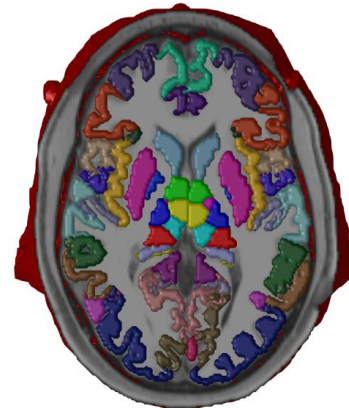
*White Matter
Volume (WM)*



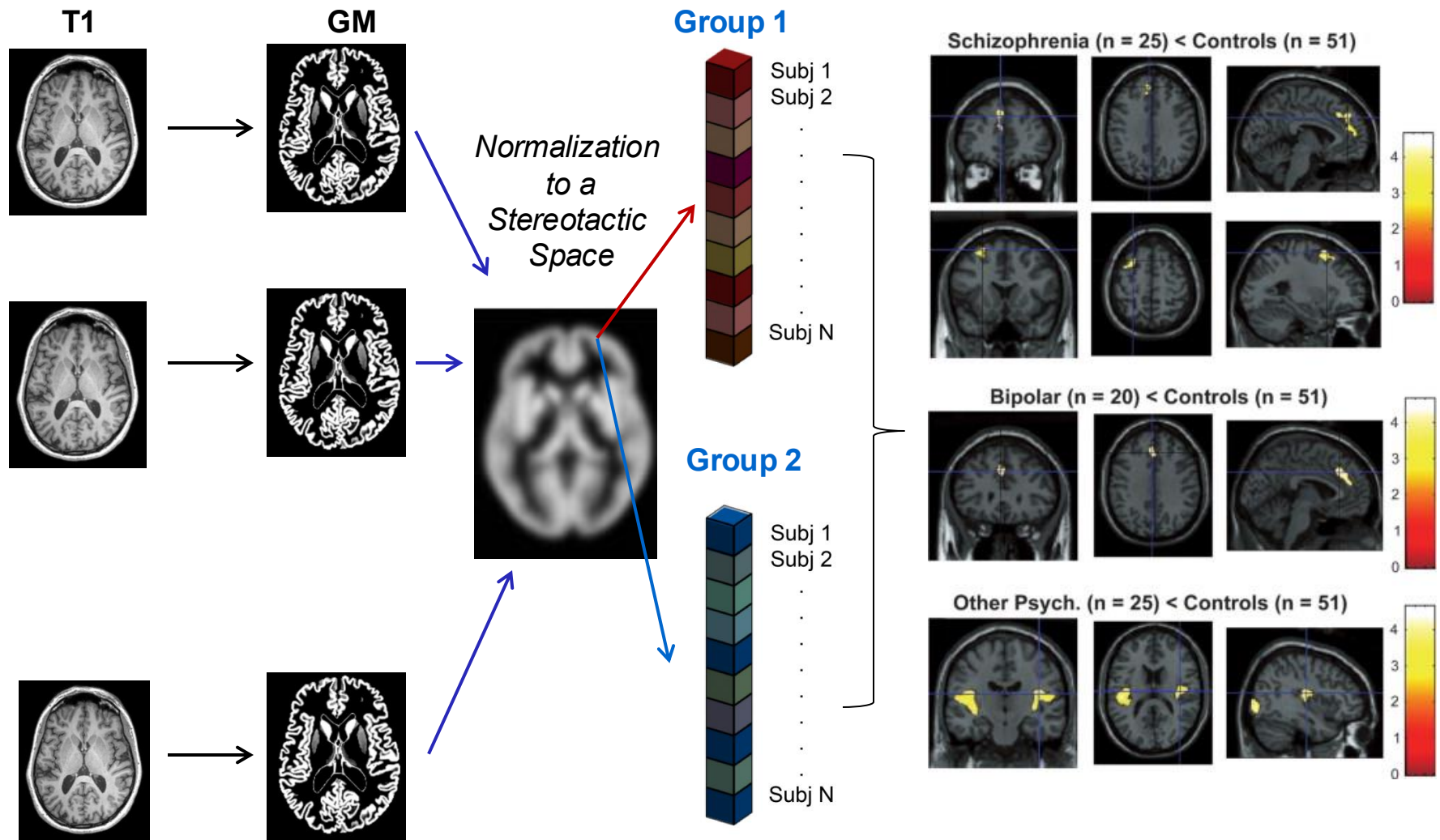
Cortical Gray Matter Volume



Anatomical parcellation



Voxel-Based Morphometry (VBM)



Voxel-Based Morphometry (VBM)

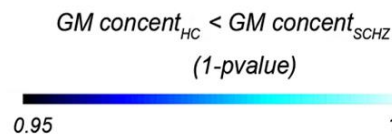
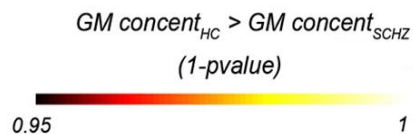
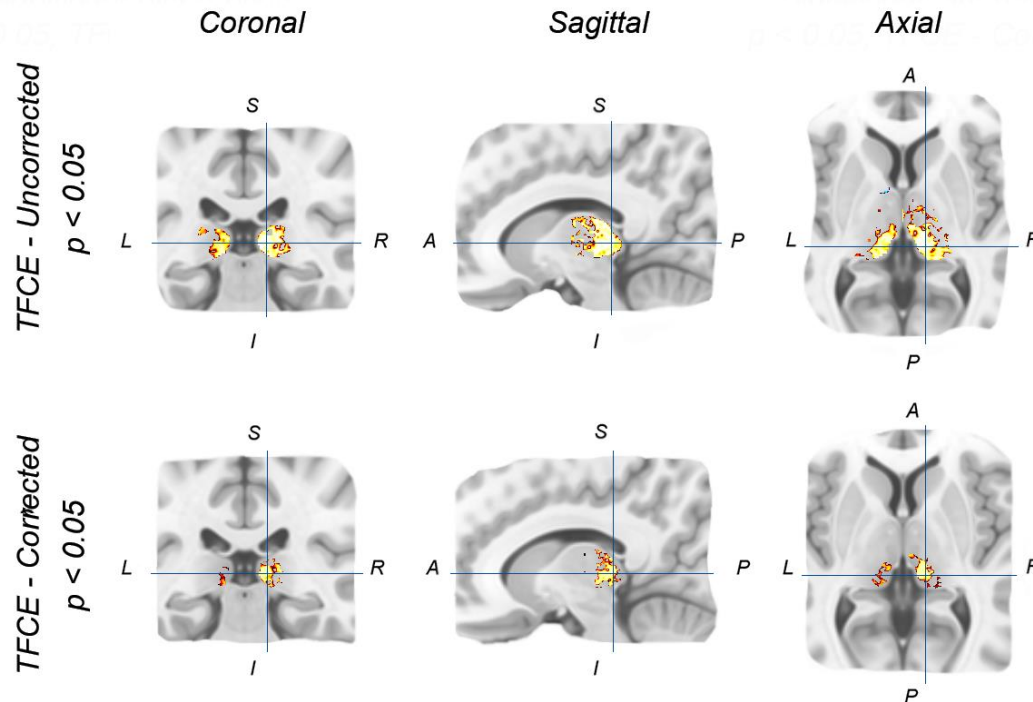
Significant differences in GM concentration between groups

Advantages

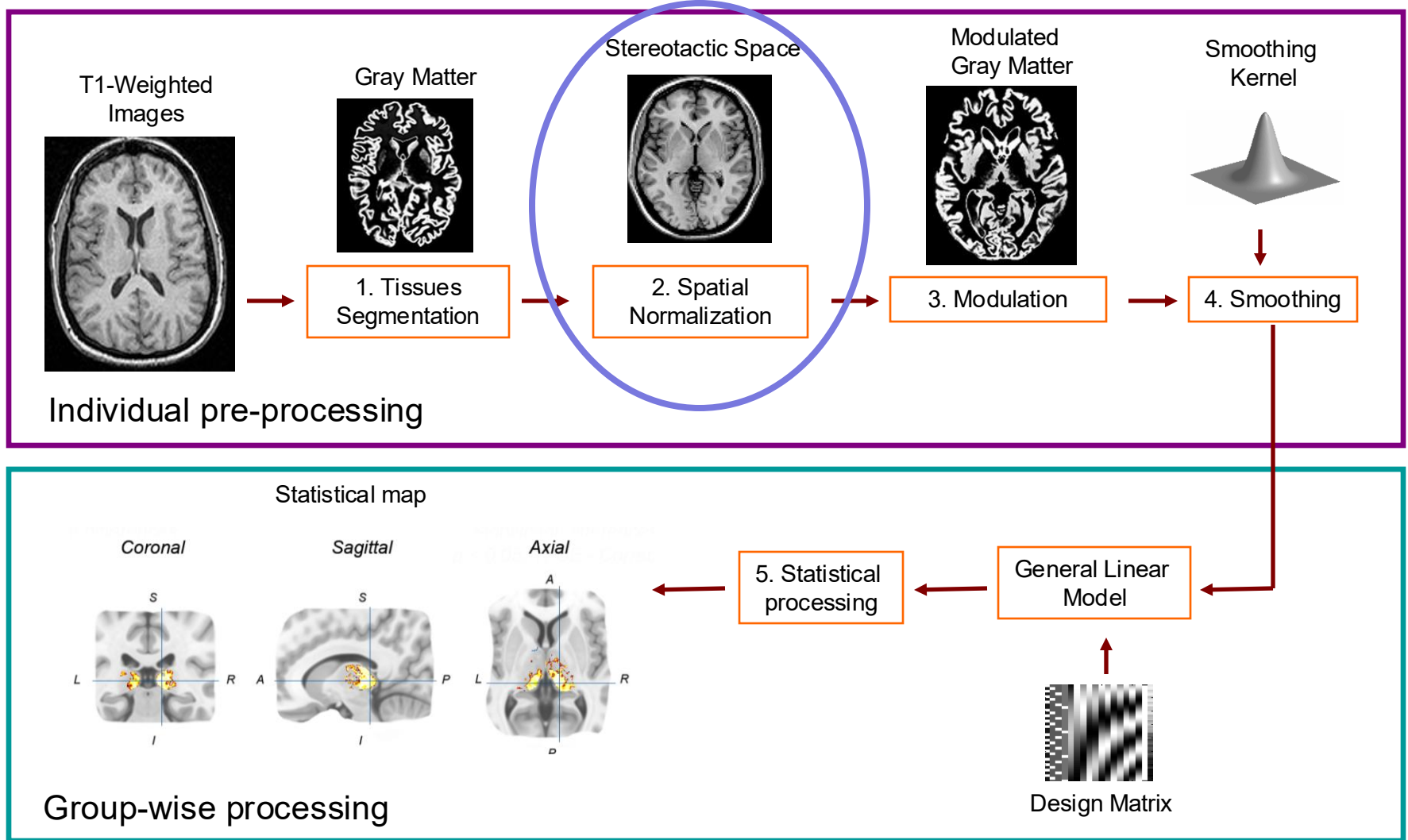
- No apriori spatial information is needed

Drawbacks

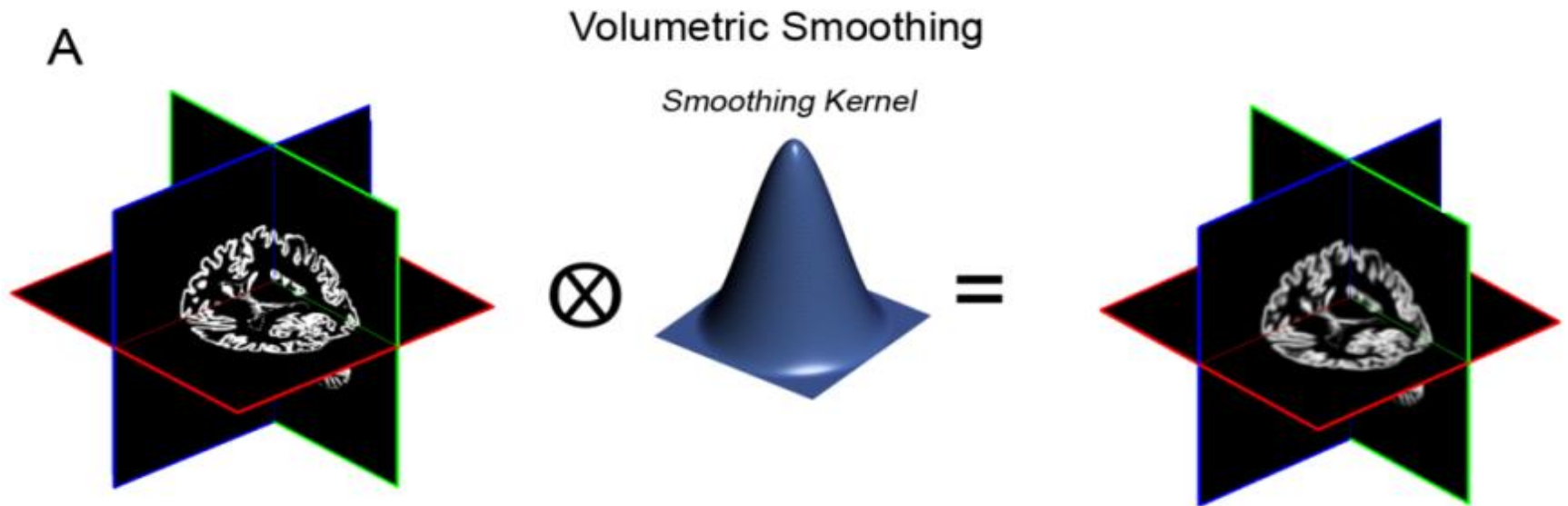
- Strong alignment between subjects in stereotactic space



Voxel-based morphometry



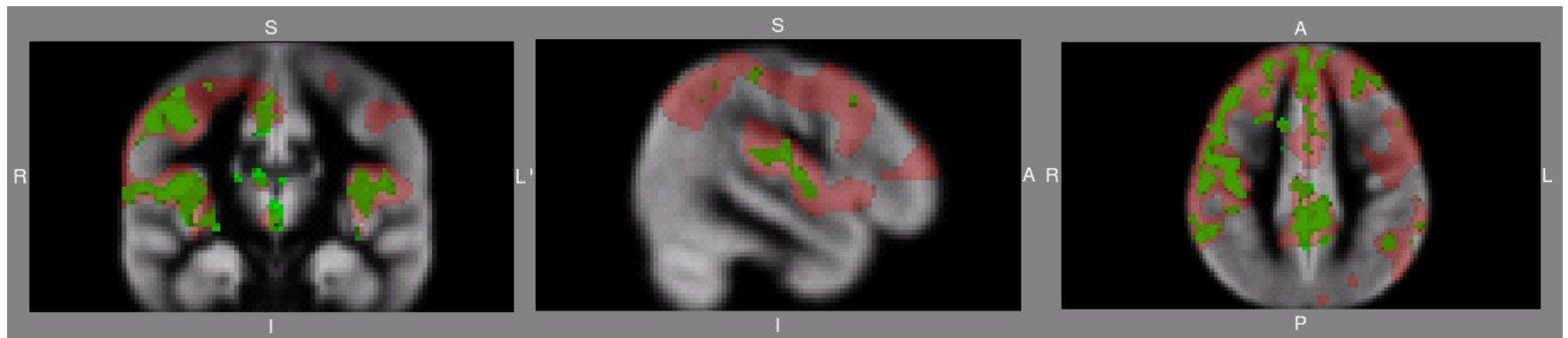
Volumetric smoothing



Smoothing

- The data will be more Gaussian and closer to a continuous random field for larger kernels
- Results will be rough and noise-like if too little smoothing is used
- Too much will lead to distributed, indistinct blobs
 - Kernel size recommended: $\geq 6\text{mm}$ and $\leq 12\text{mm}$

Results for different smoothing Kernels



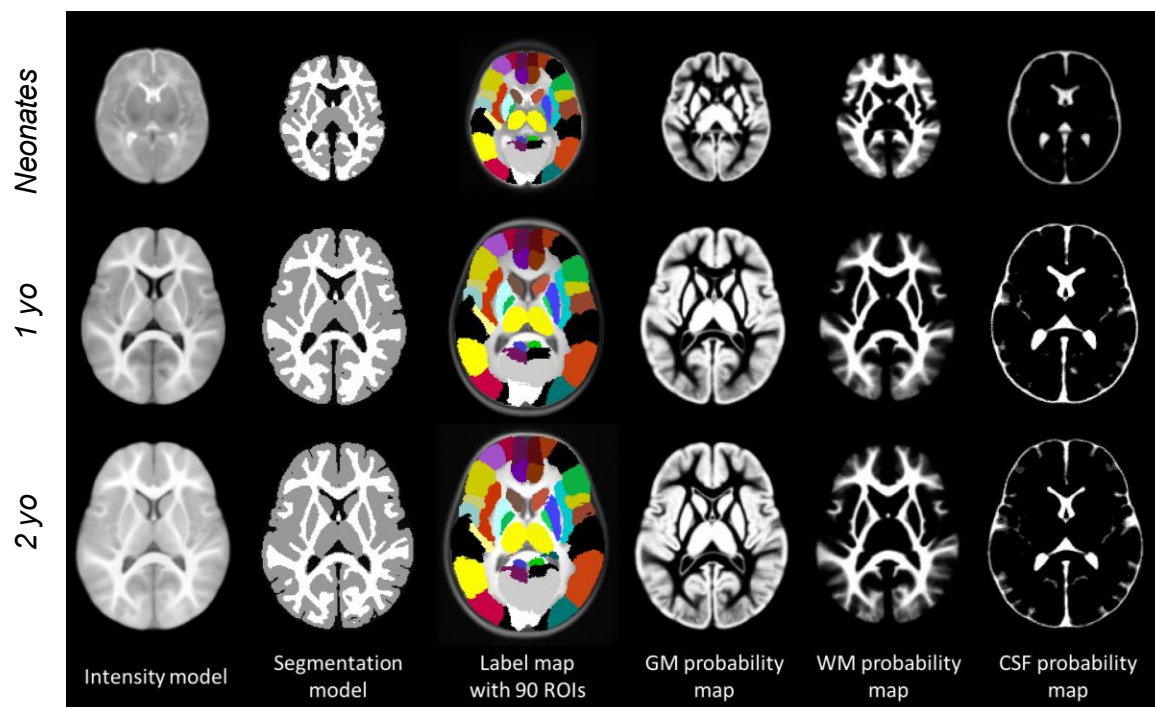
Green: FWHM = 8mm

Red: FWHM = 12mm

Population-specific template

- Custom template building

- Tissues contrast
- Age range
- Gender
- Ethnic



- Symmetric group-wise diffeomorphic normalization (SyN) ^[1] and FNIRT

ANTs

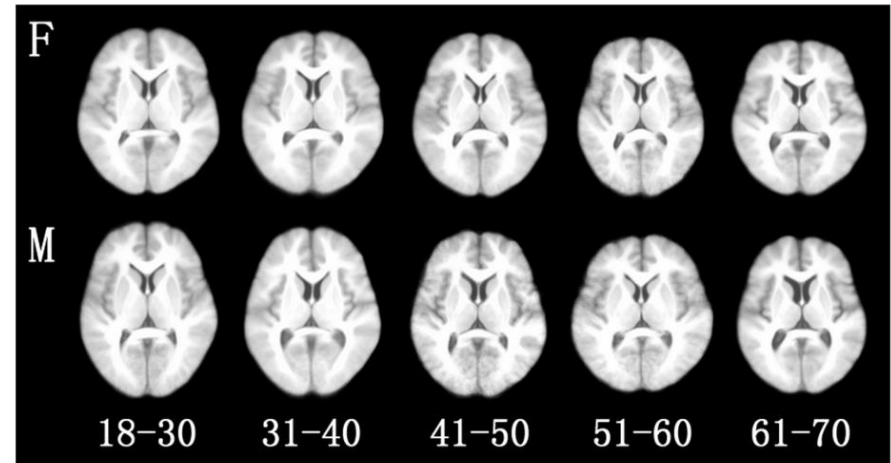
(*buildtemplateparallel.sh*, C++)

FSL

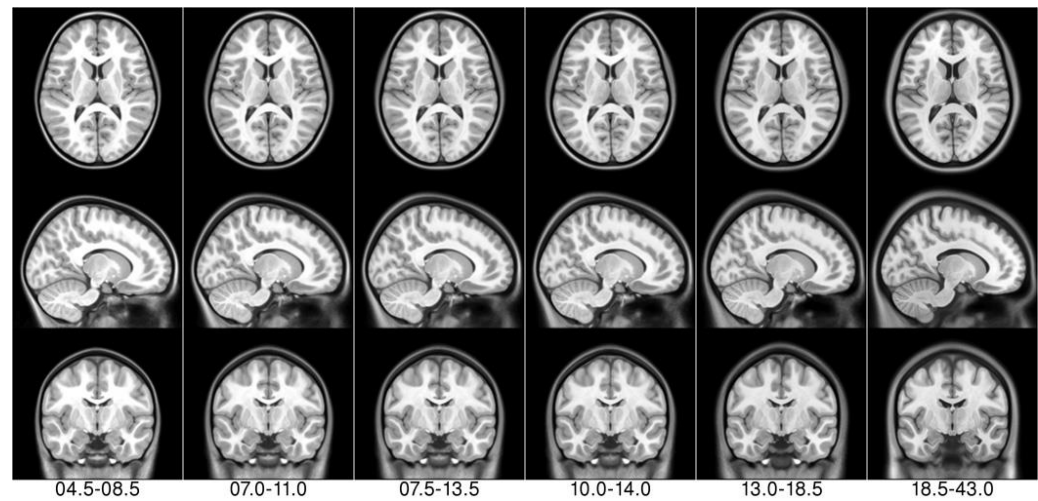
(*fslvbm_2_template*, C++)

Population-specific template

*Probabilistic MRI Brain Anatomical Atlases
Based on 1000 Chinese Subjects [1]*



*Unbiased Average Age-Appropriate
Atlases for Pediatric Studies [2]*



[1] Xing et al 2013. Plos One

[2] Fonov et al 2011. Neuroimage

Interpretation of VBM results

Reference



Mis-classification



Mis-registration



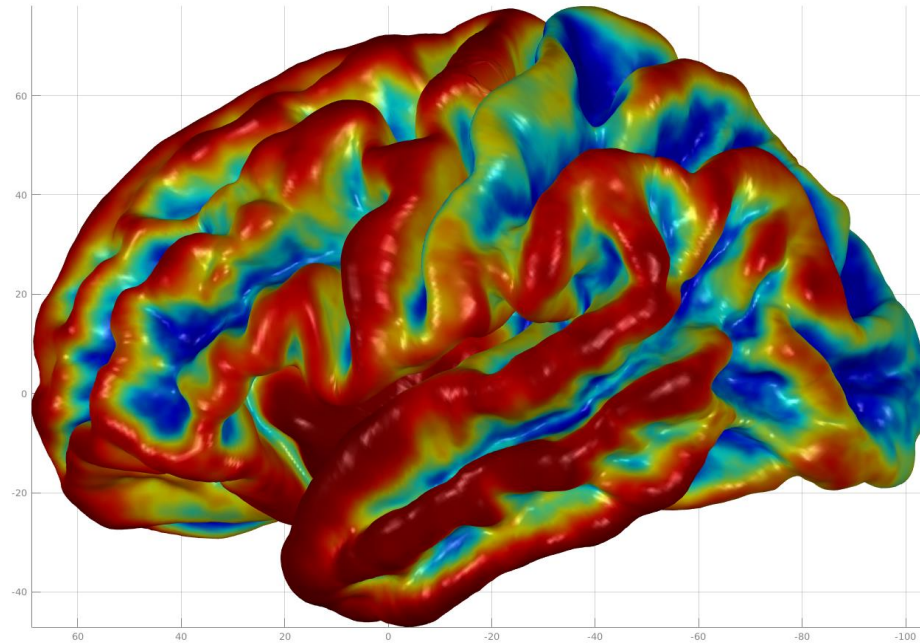
Thickening or thinning



Unfolding



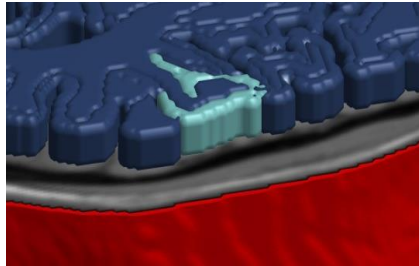
- In essence VBM is Statistical Parametric Mapping of regional segmented tissue density or volume
- The exact interpretation of gray matter density or volume is complicated, and depends on the preprocessing steps used
 - It is not interpretable as neuronal packing density or other cytoarchitectonic tissue properties
 - The hope is that changes in these microscopic properties may lead to macro- or mesoscopic VBM-detectable differences



Surface Based Morphometry (SBM)

Volume as a composite measure

Cortical region



Box



Pyramid

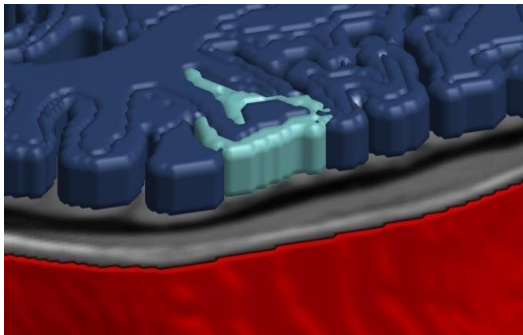


Teapot

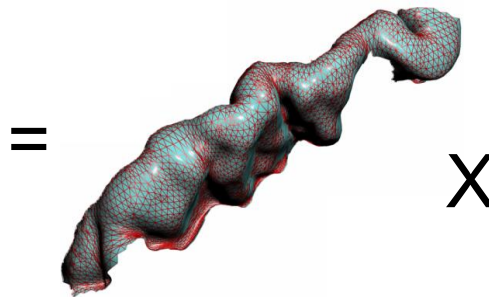


Same volume different shape

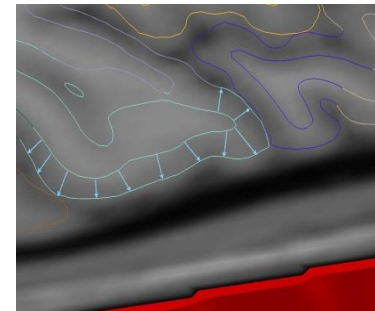
Volume



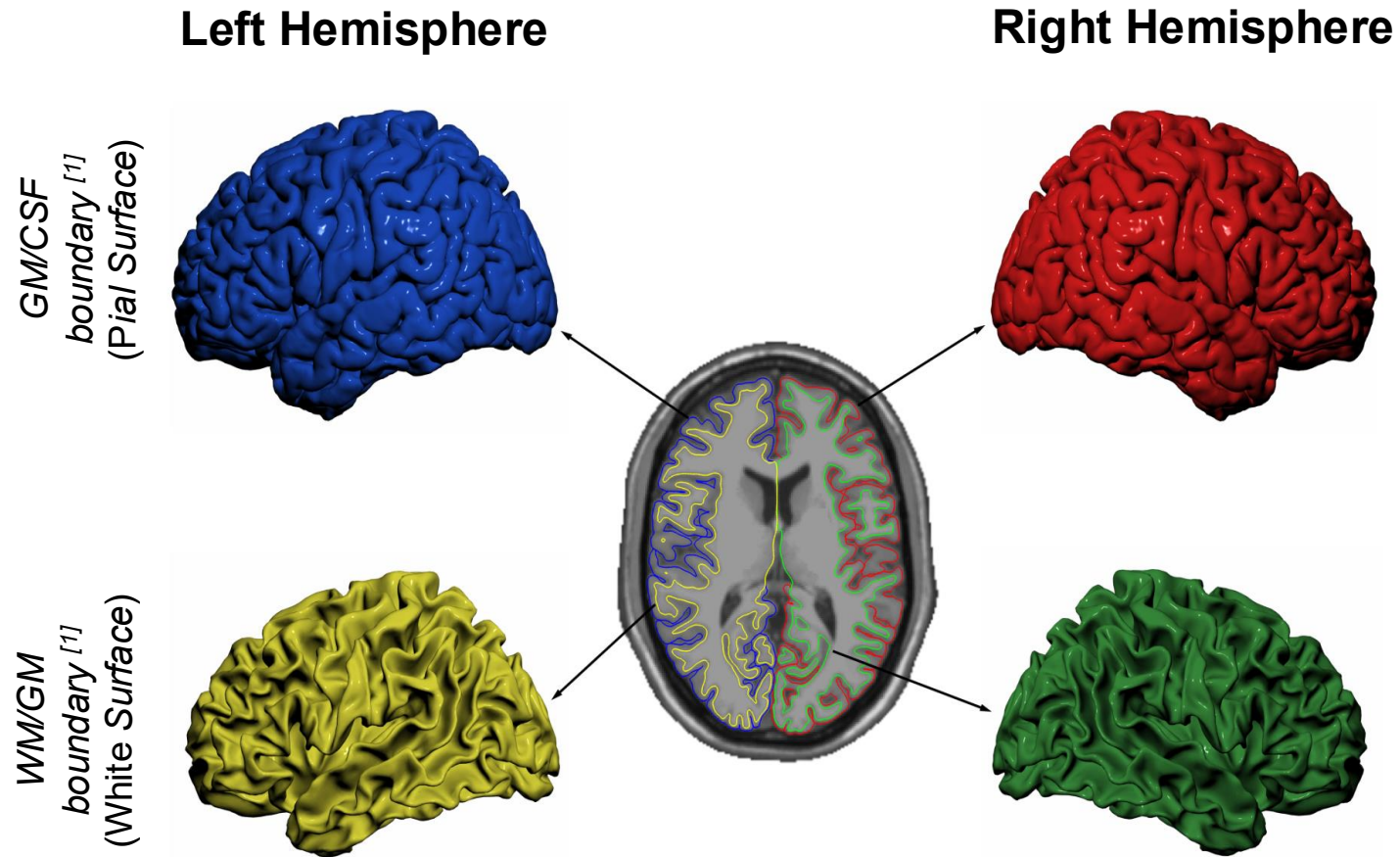
Surface Area



Cortical thickness



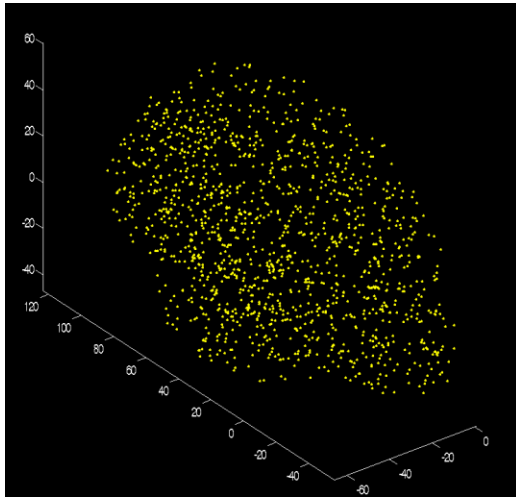
Cortical surfaces



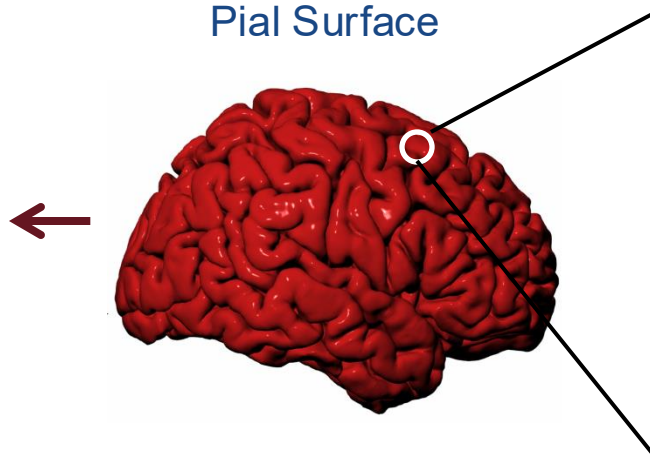
[1] Davatzikos et al. 1996. *IEEE Transactions on Medical Imaging*

Basic concepts:

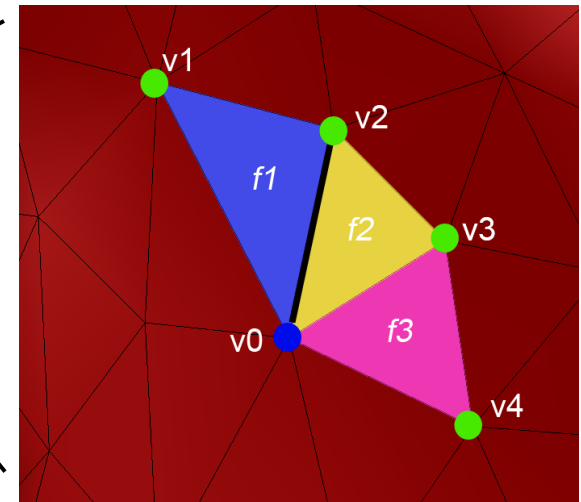
Pial Surface vertices



Pial Surface



Faces



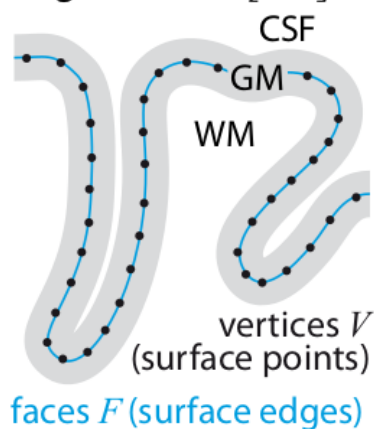
$$f1 = (v0, v1, v2)$$

$$f2 = (v0, v2, v3)$$

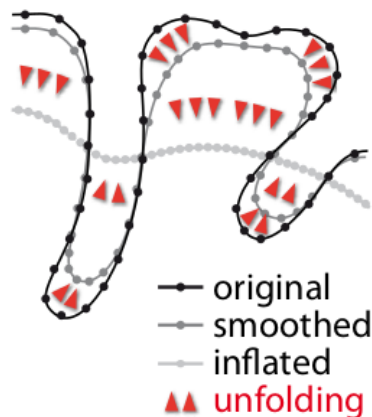
$$f3 = (v0, v3, v4)$$

Operations with surfaces

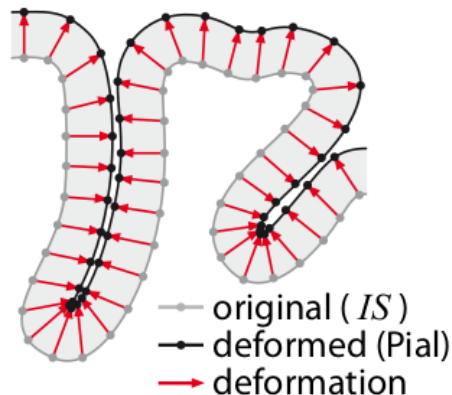
original $CS = [V, F]$



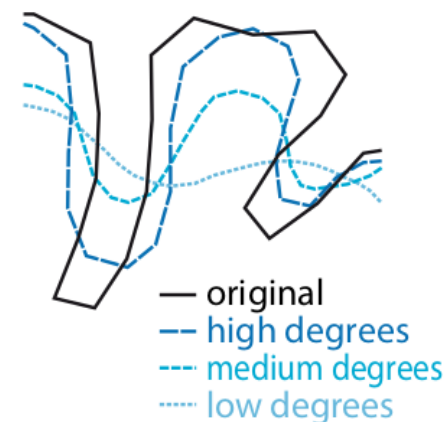
a) Smoothing & Inflation



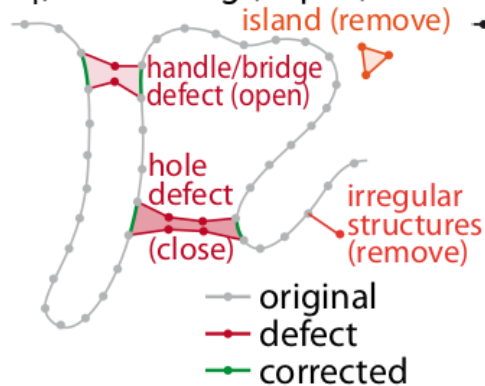
b) Deformation



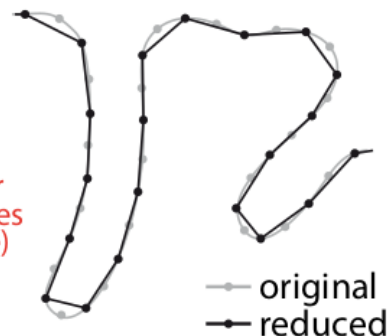
d) Parameterization



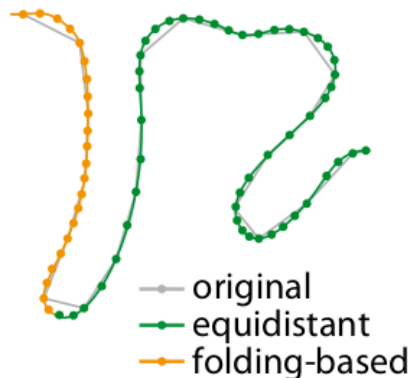
c₁) Remeshing (Repair)



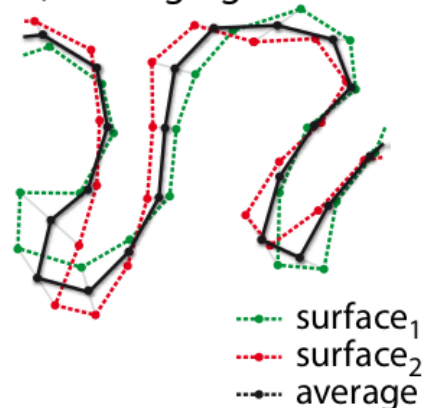
c₂) Remeshing (Reduction)



c₃) Remeshing (Refinement)

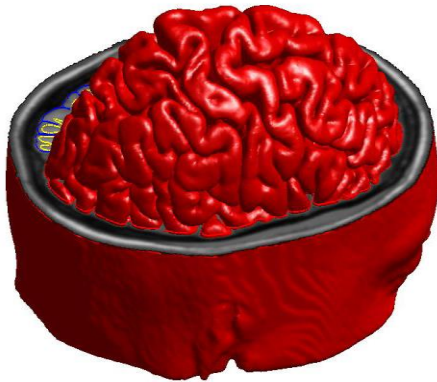


e) Averaging

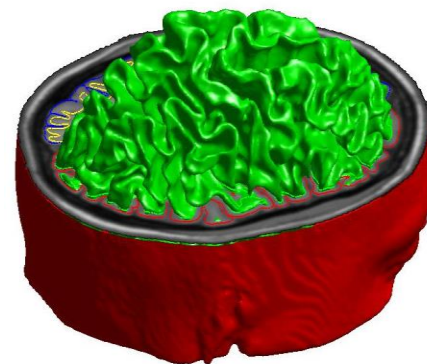


Cortical surfaces

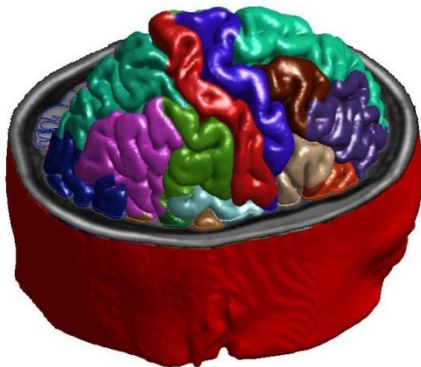
**GM/CSF
(Pial Surface)**



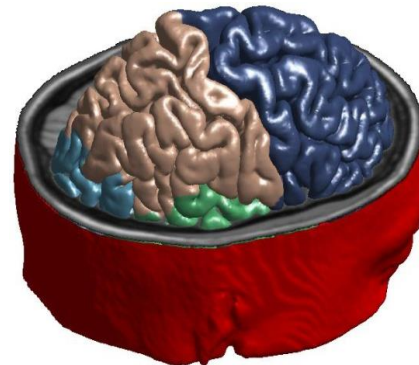
**WM/GM
(White Surface)**



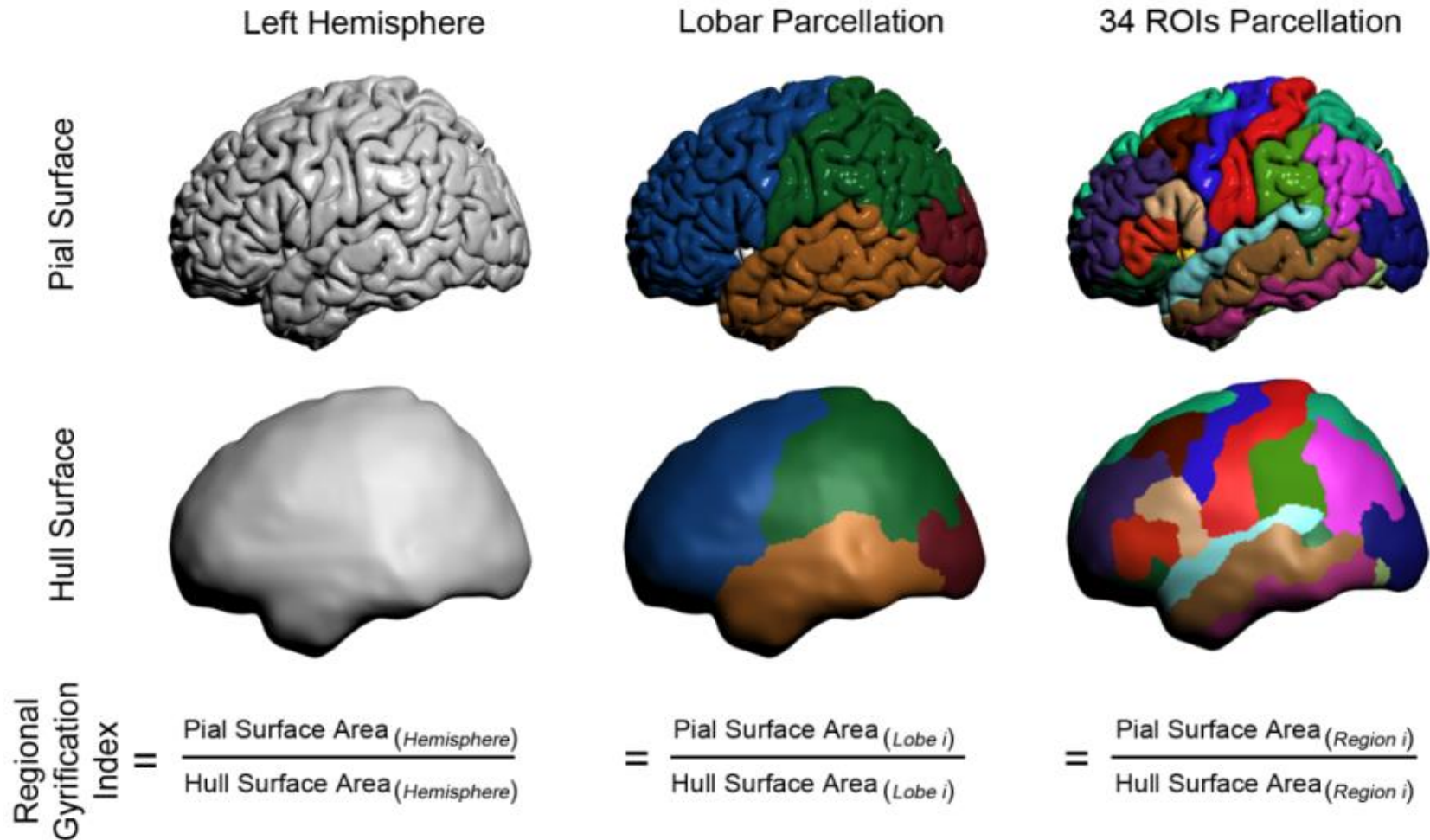
**Cortical parcellation
(Desikan et al 2006)**



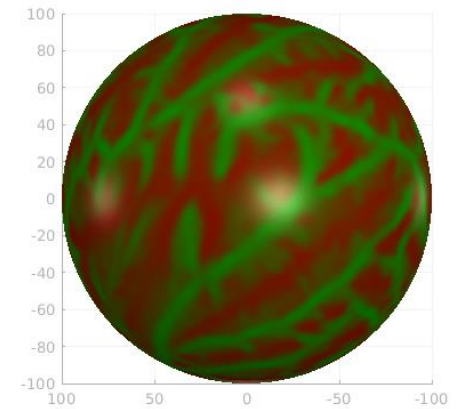
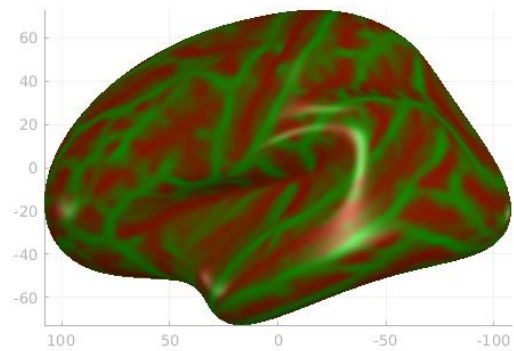
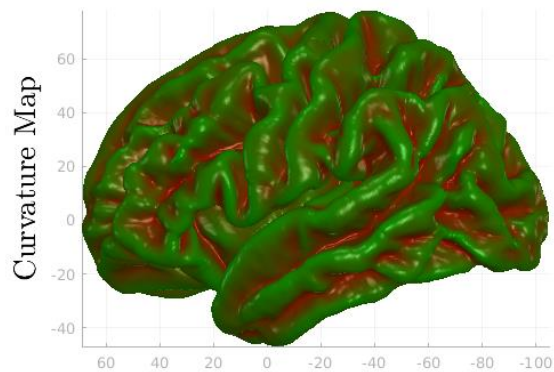
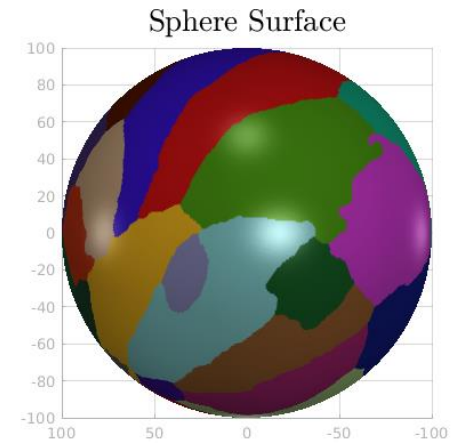
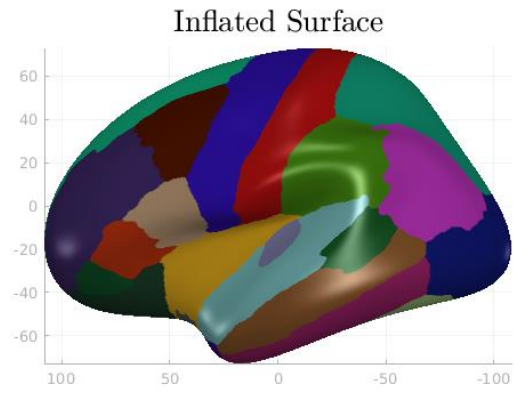
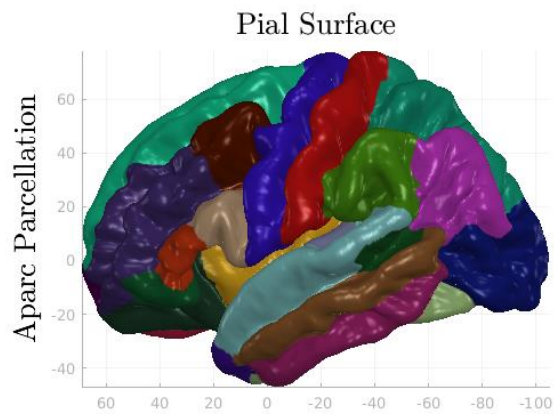
**Lobar
parcellation**



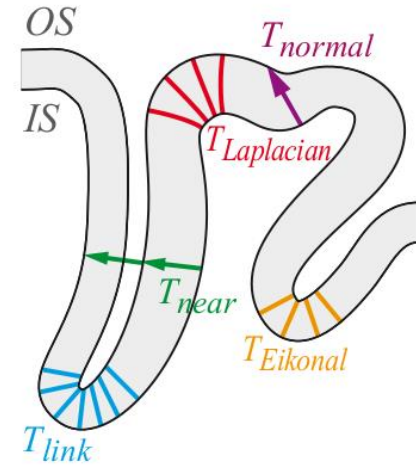
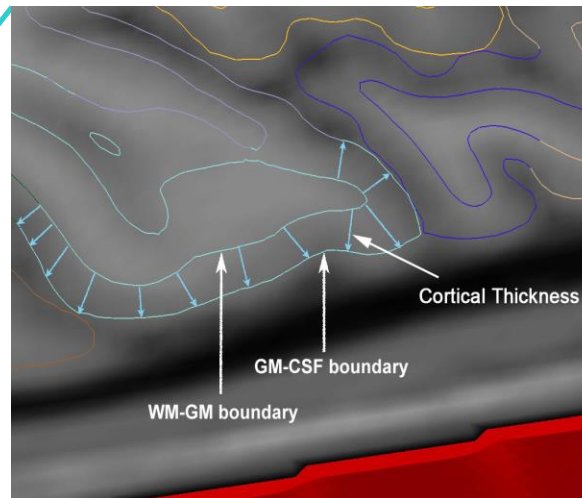
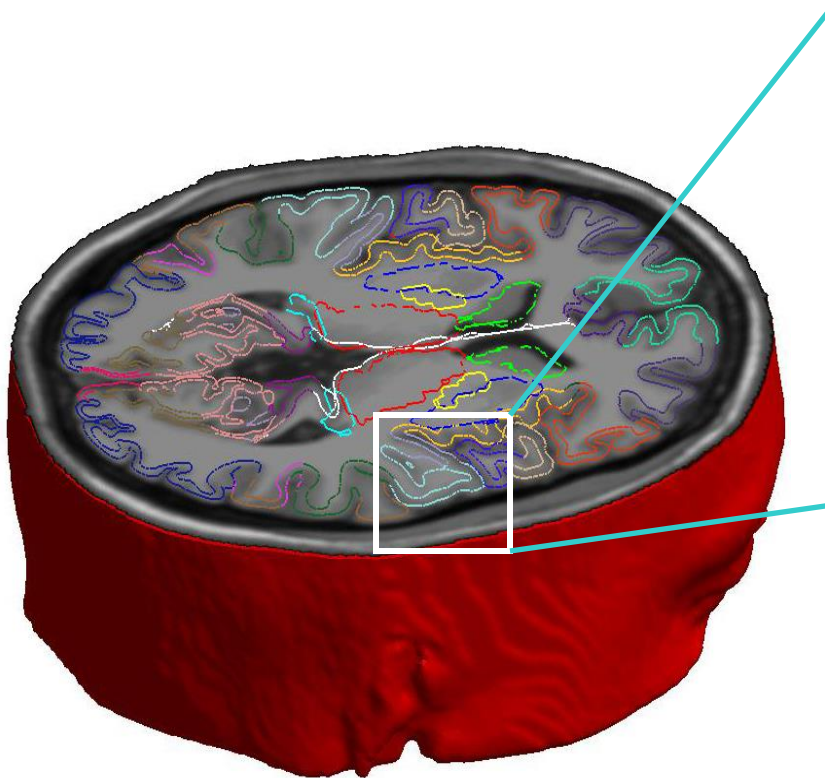
Regional metrics



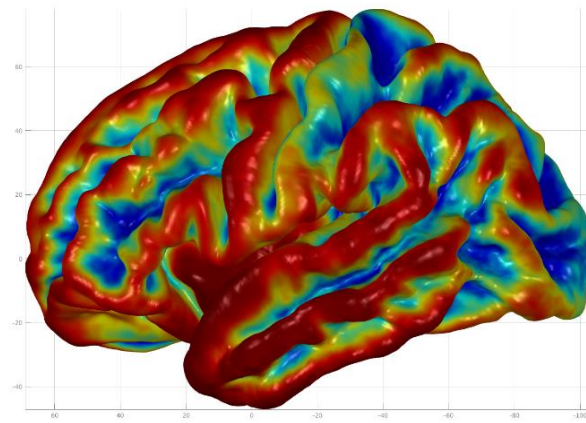
Useful surfaces



Cortical thickness

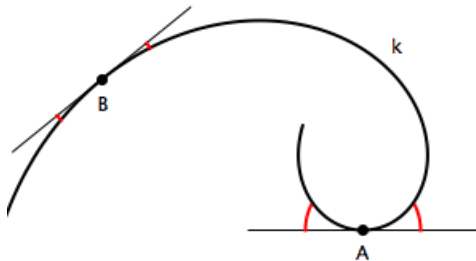


Cortical thickness map

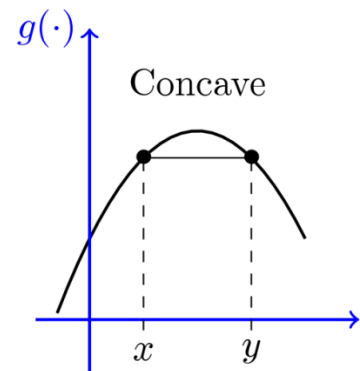
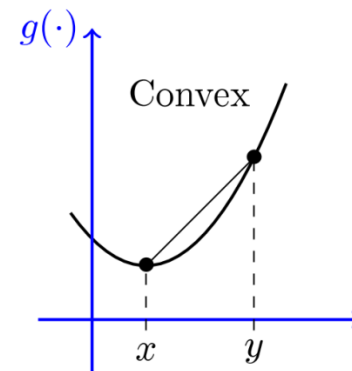


Curvature maps

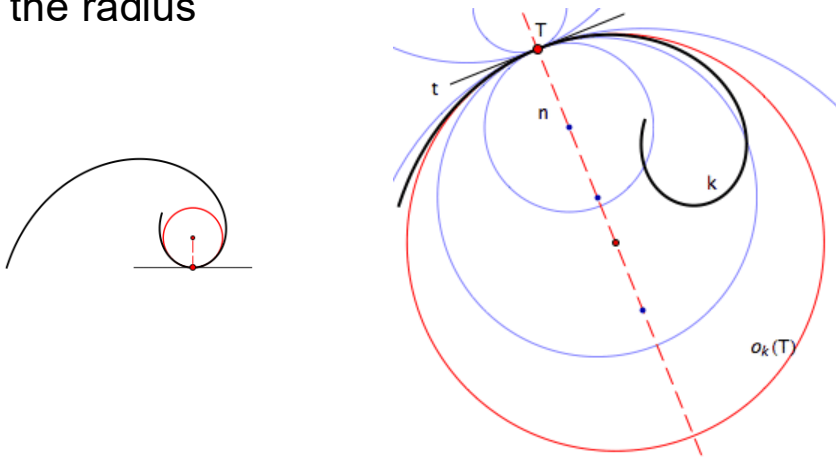
Curvature in 2D



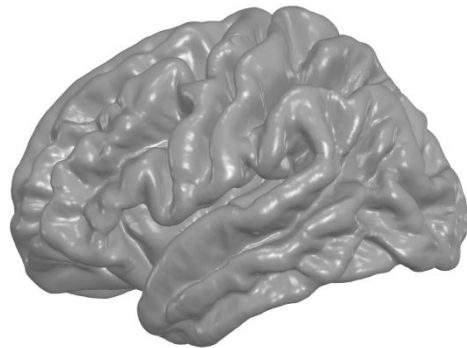
Convex vs concave curves



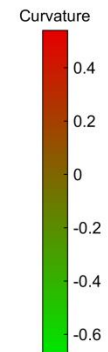
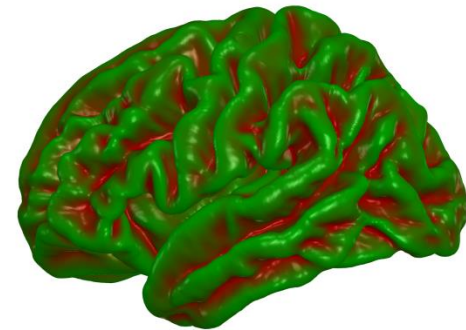
Curvature is usually defined as the inverse of the radius



Curvature maps and sulcal basins



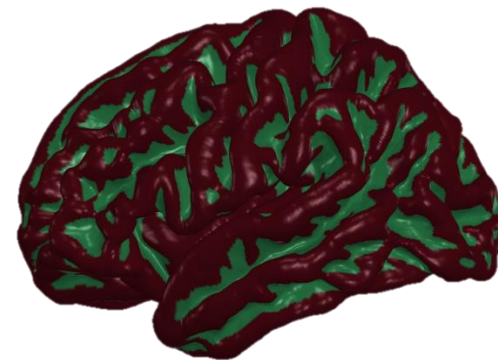
Pial cortical Surface



Curvature Map



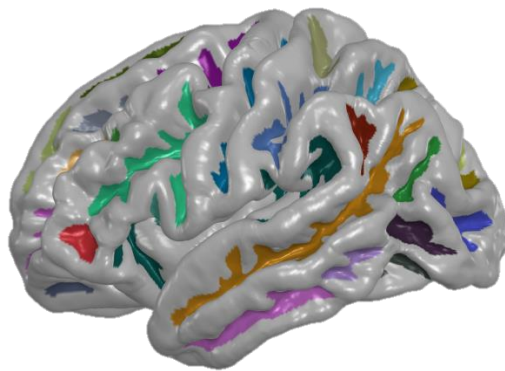
Thresholding



Green Regions: Sulcal Basins

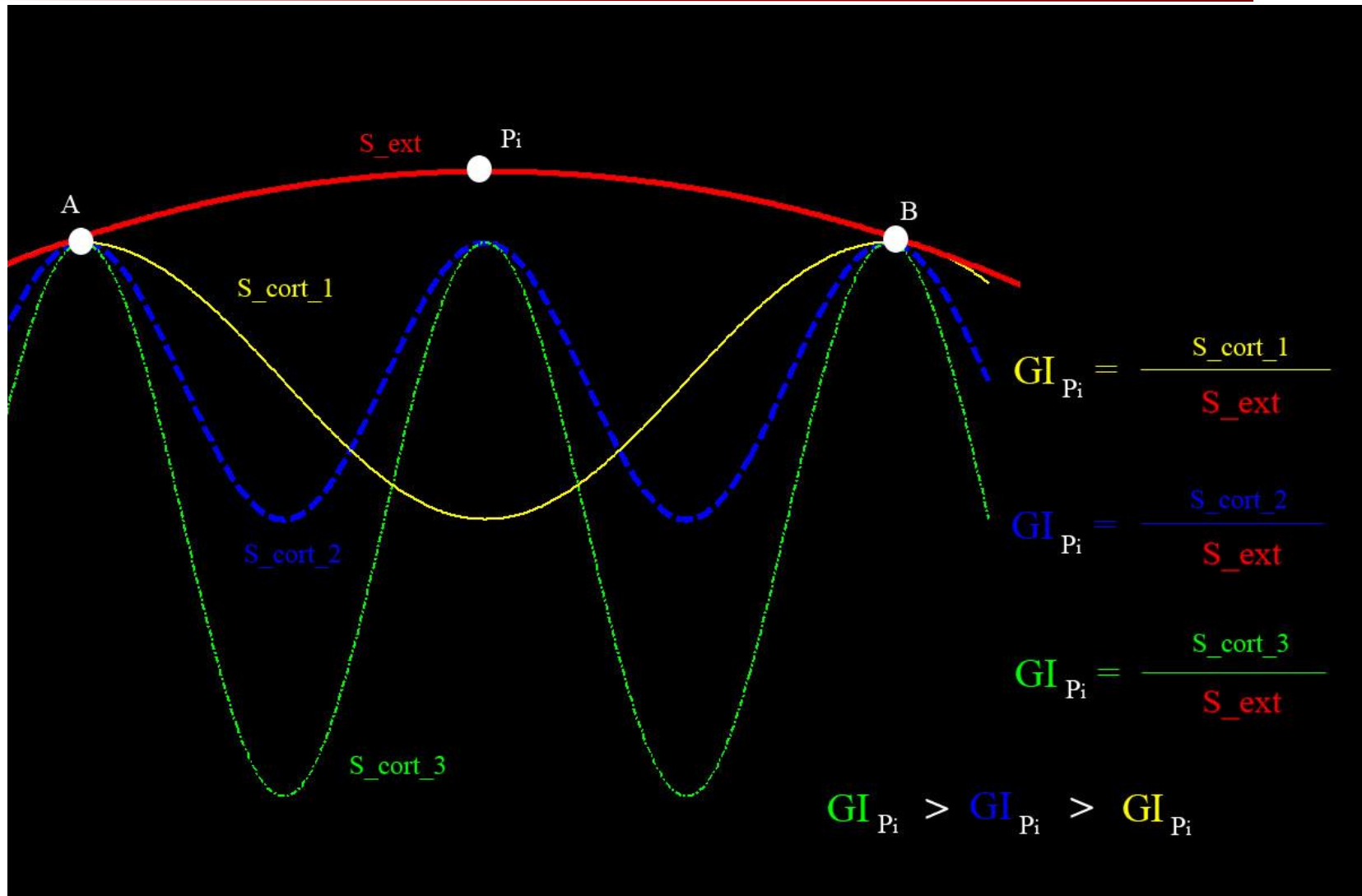


Labelling

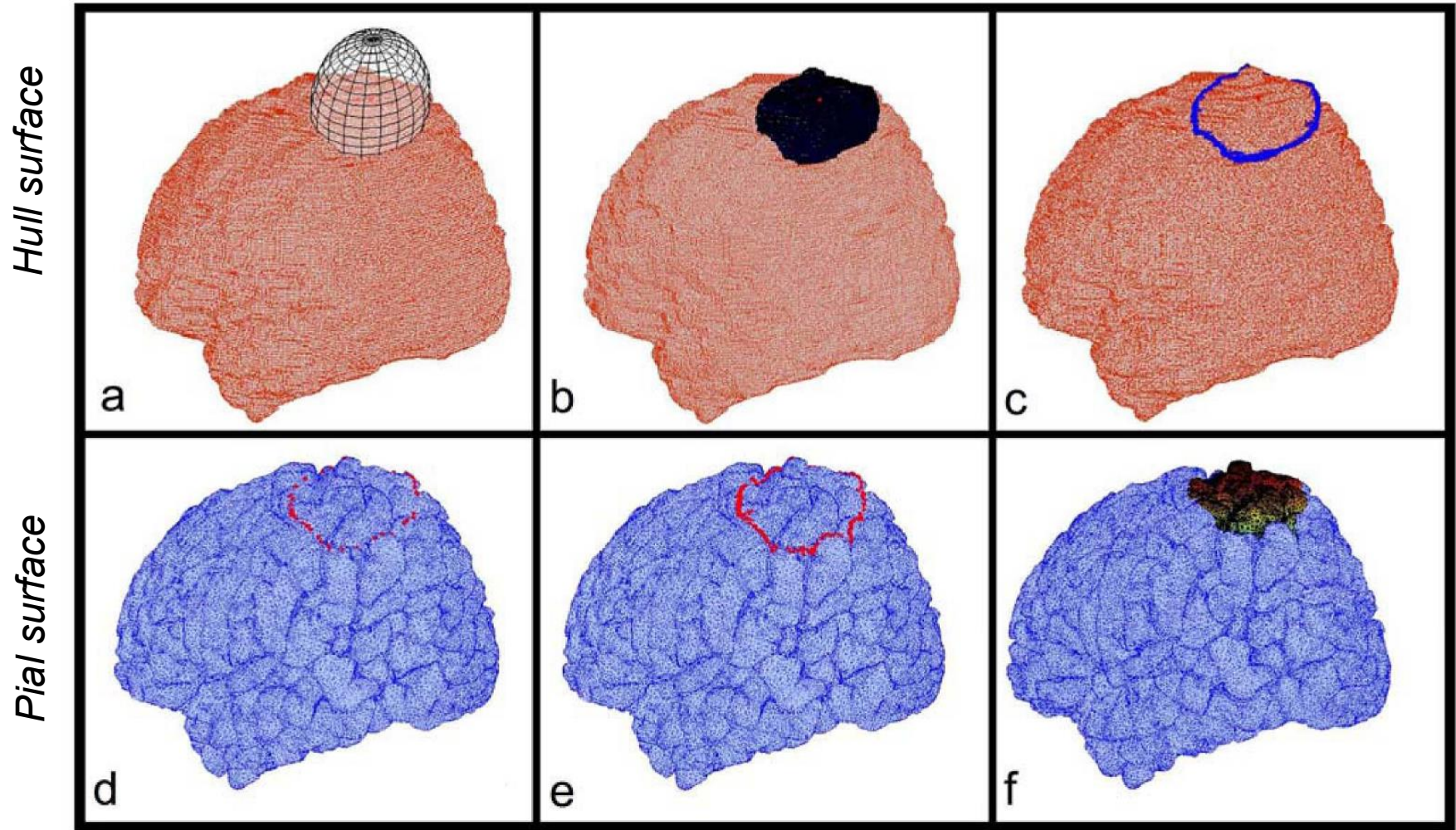


Sulcal Basins (Pial Surface)

Local gyrification index

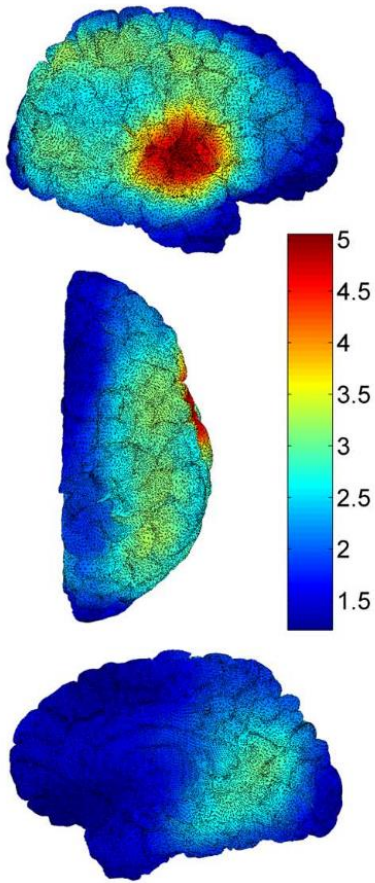


Local gyrification index

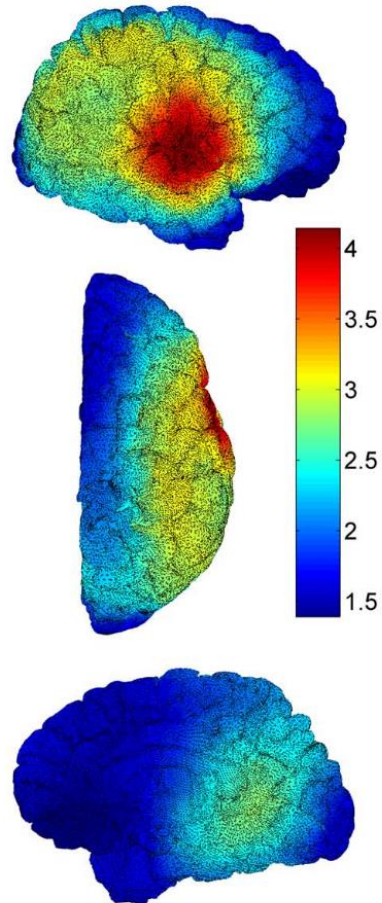


Local gyrification index

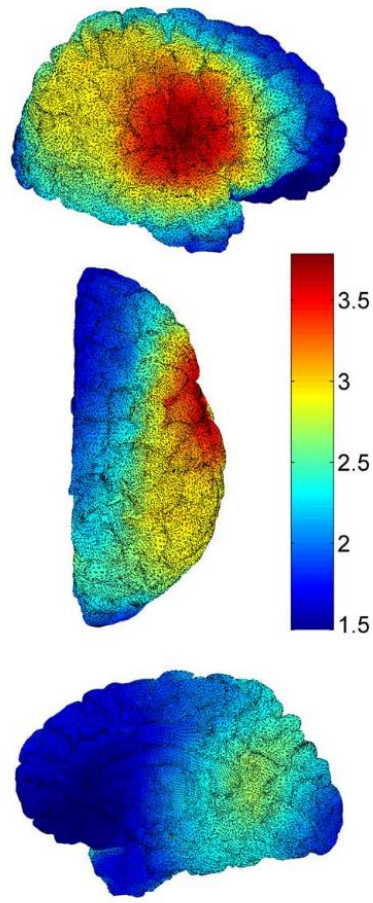
Radius = 20 mm



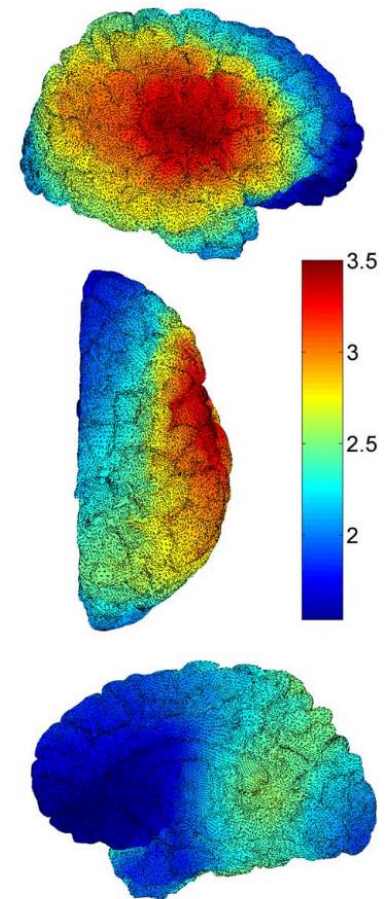
Radius = 25 mm



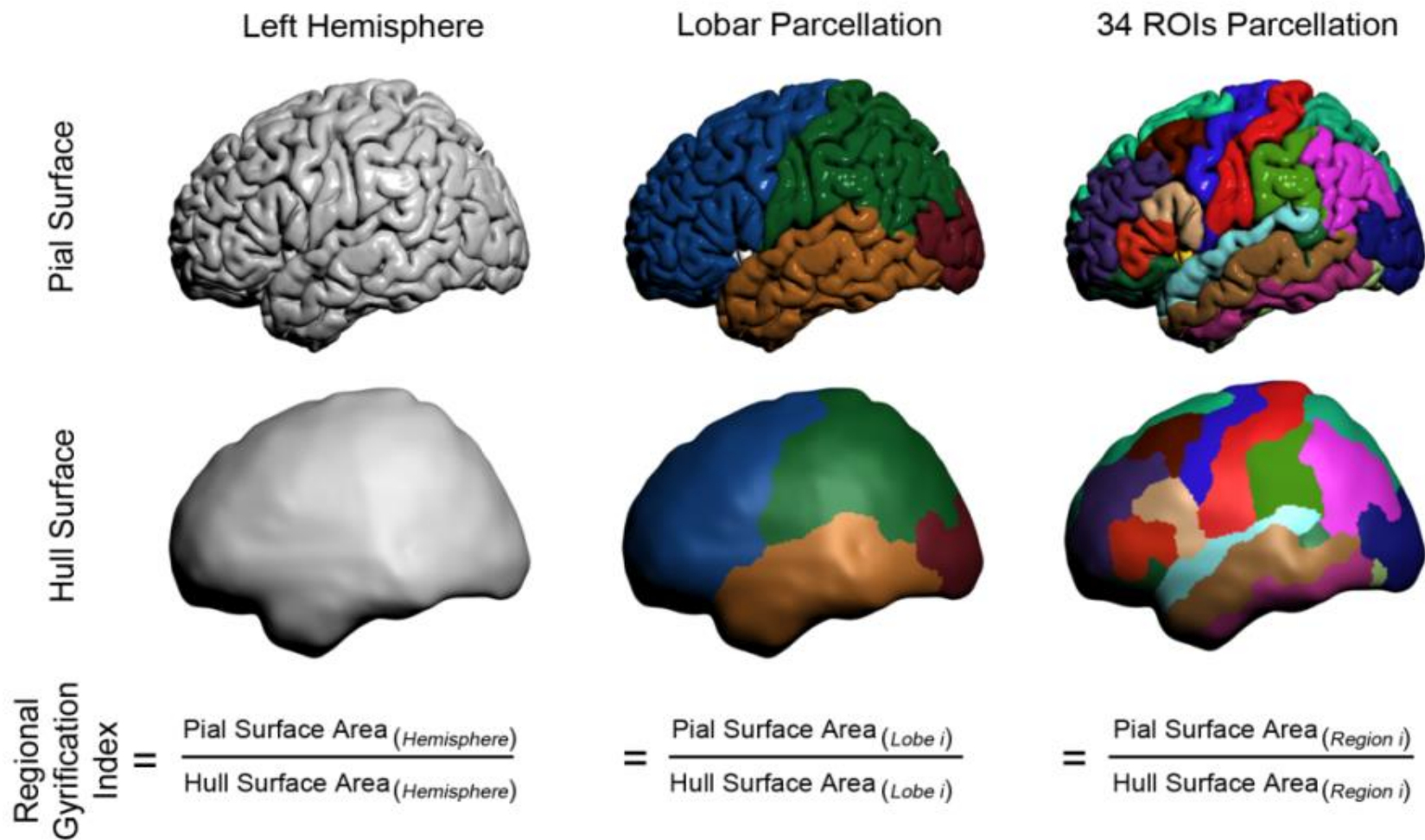
Radius = 30 mm



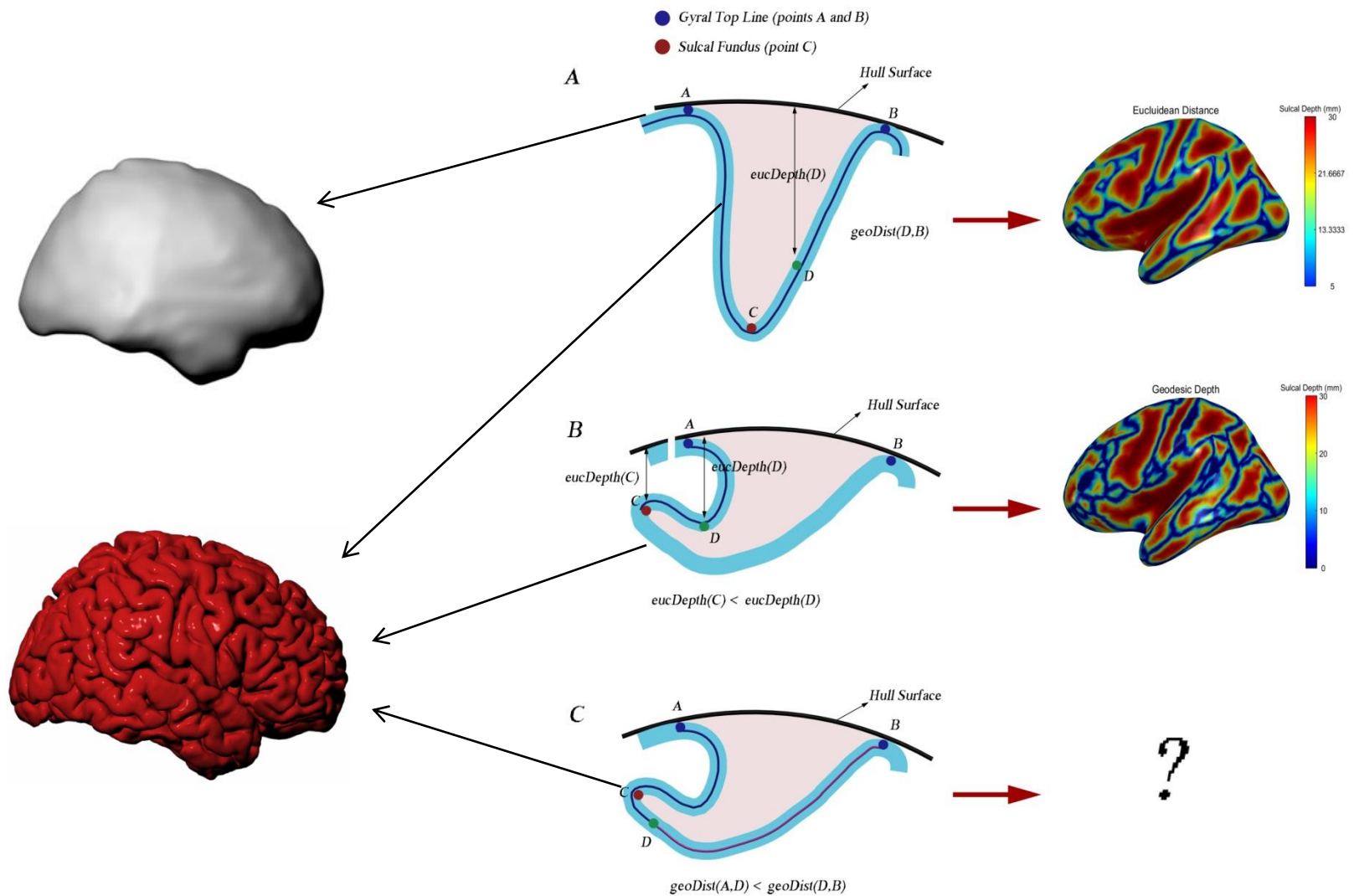
Radius = 35 mm



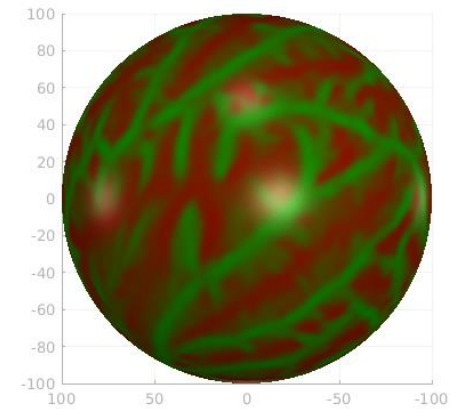
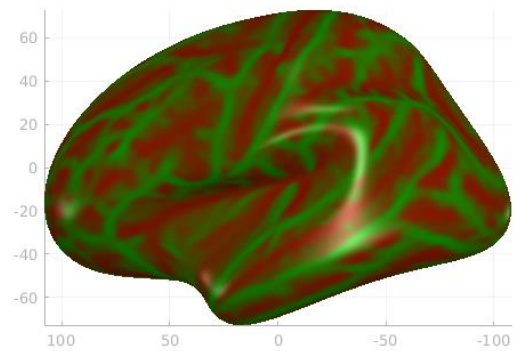
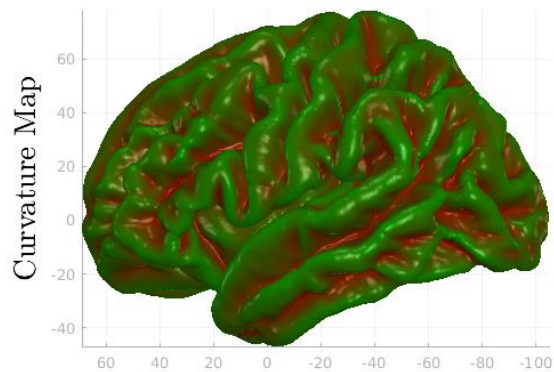
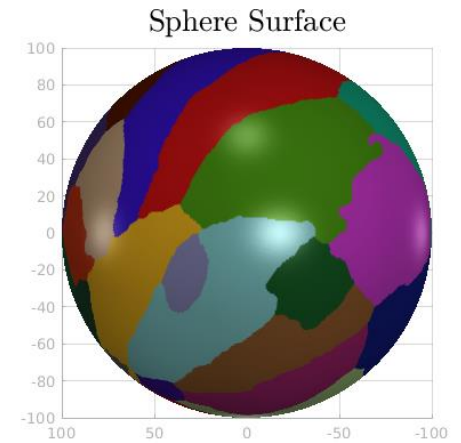
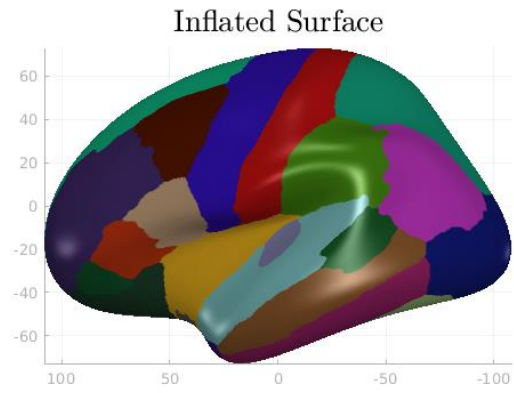
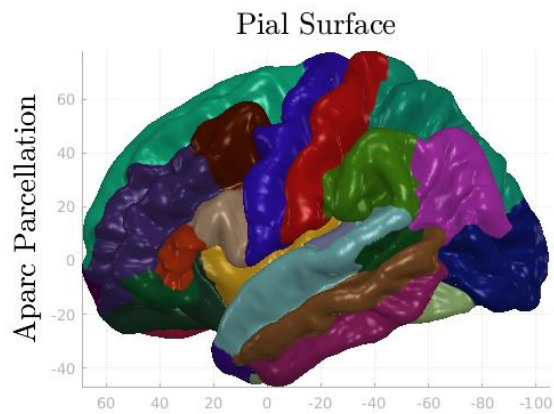
Regional gyrification index



Sulcal depth

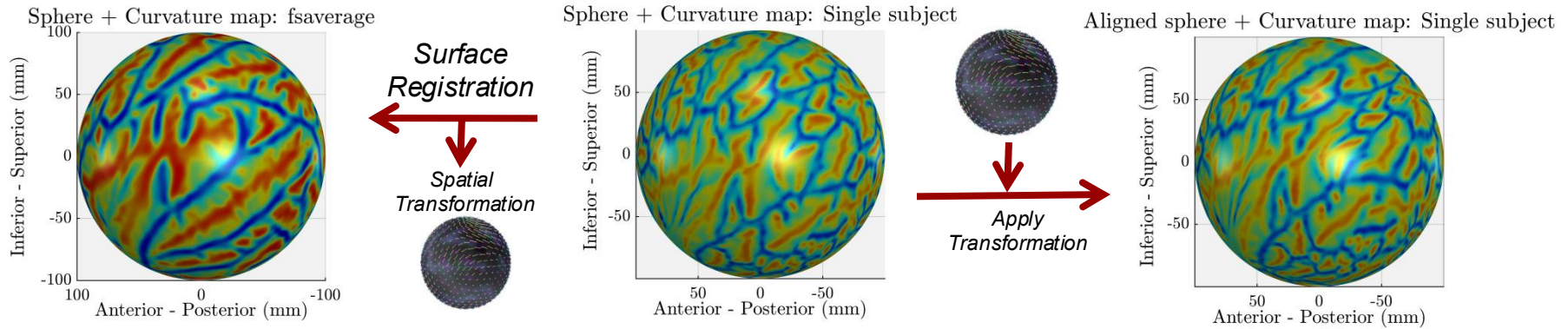


Surface mapping

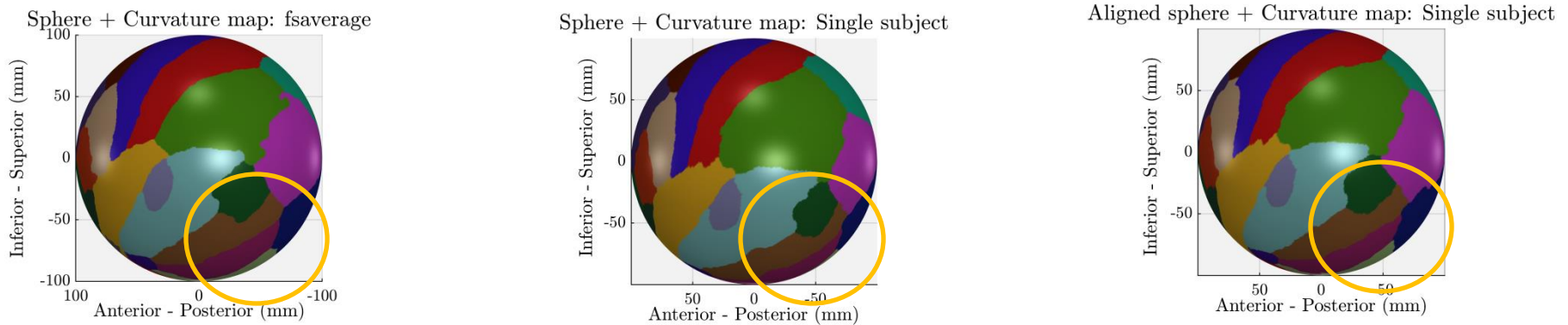


Surface alignment

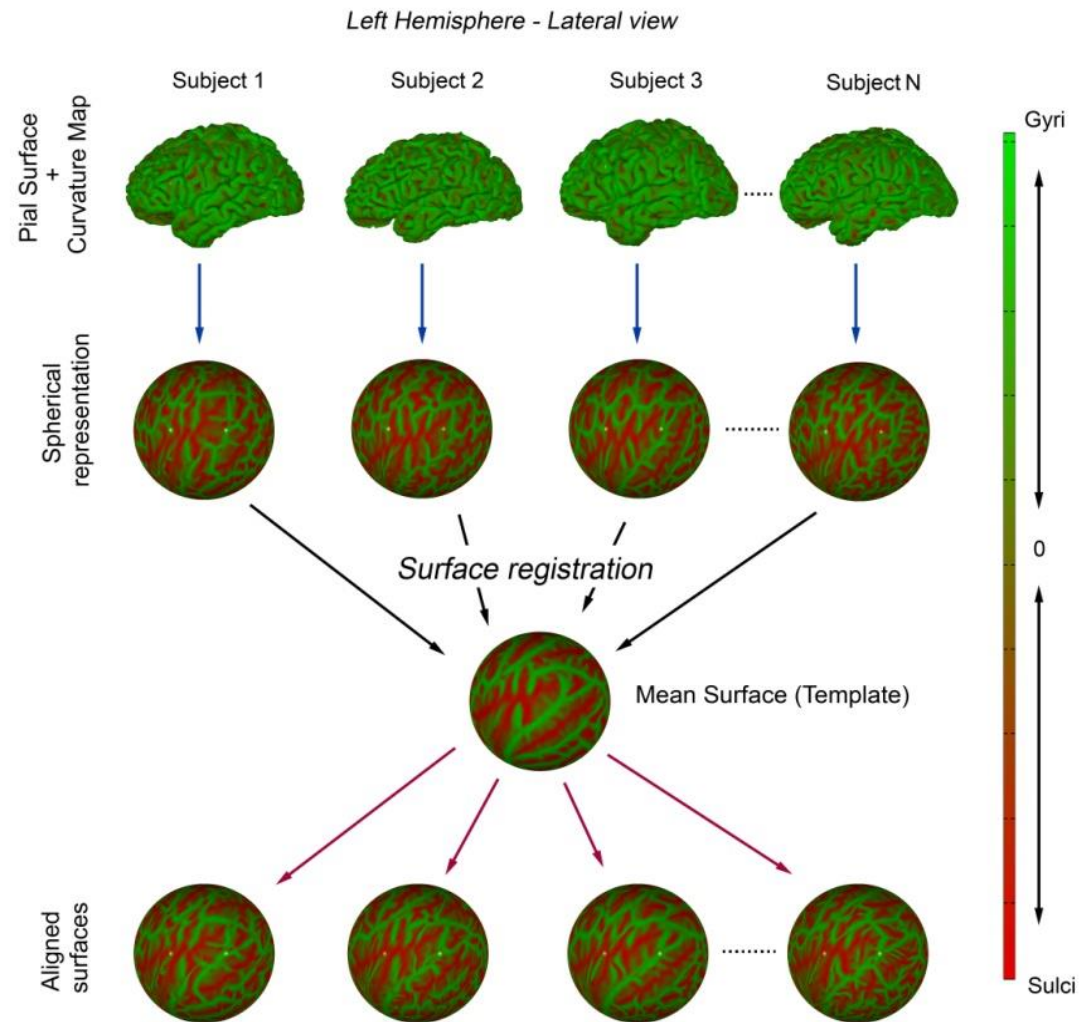
Surface alignment workflow



Cortical parcellation over the sphere



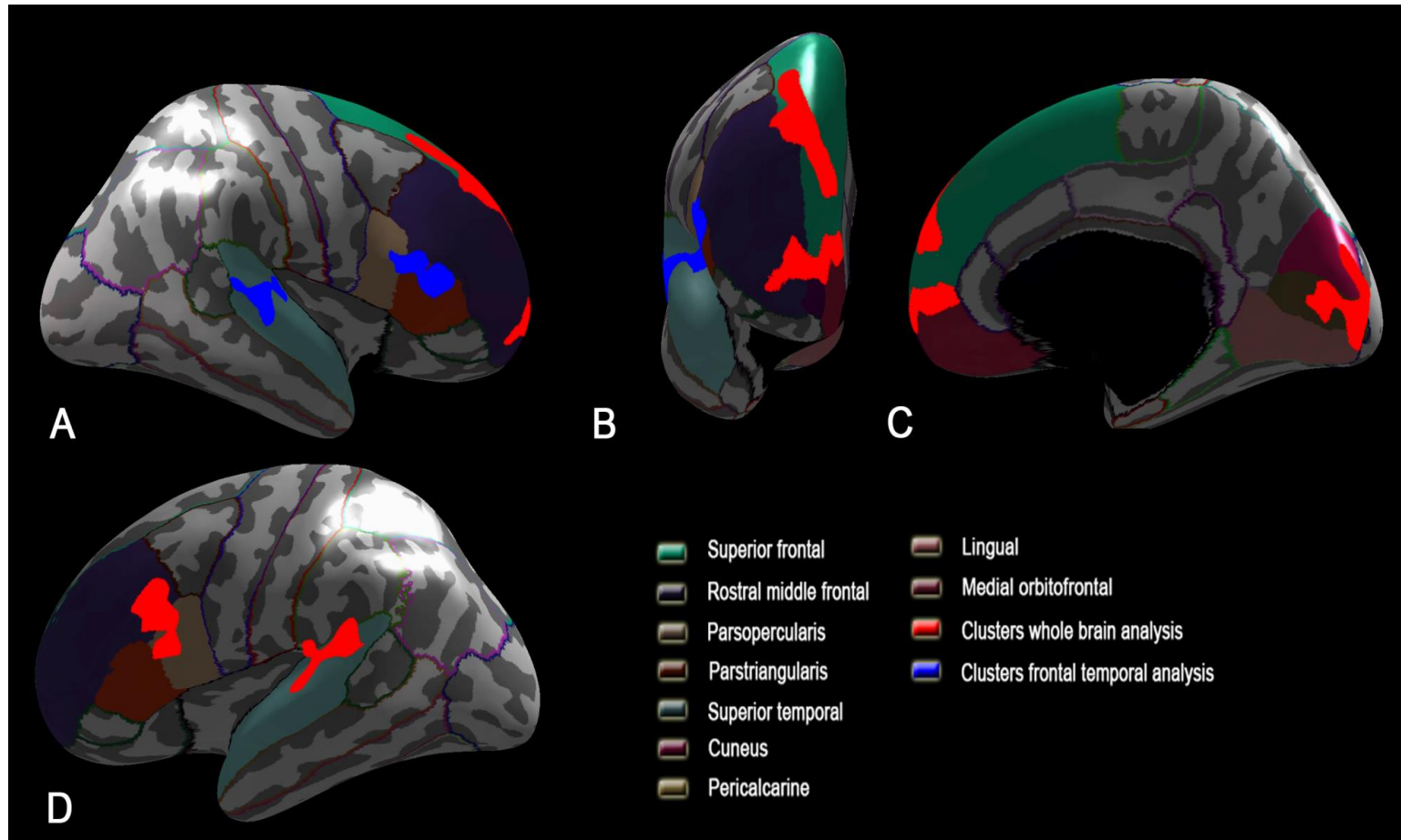
Group alignment

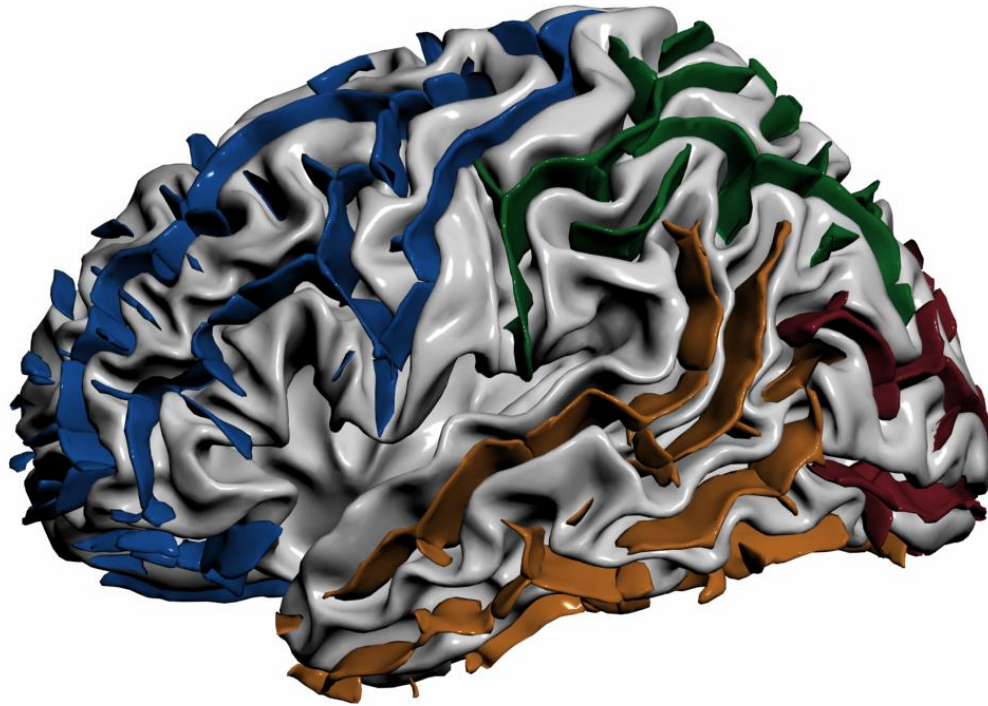


Geodesic smoothing



Surface-Based Morphometry results

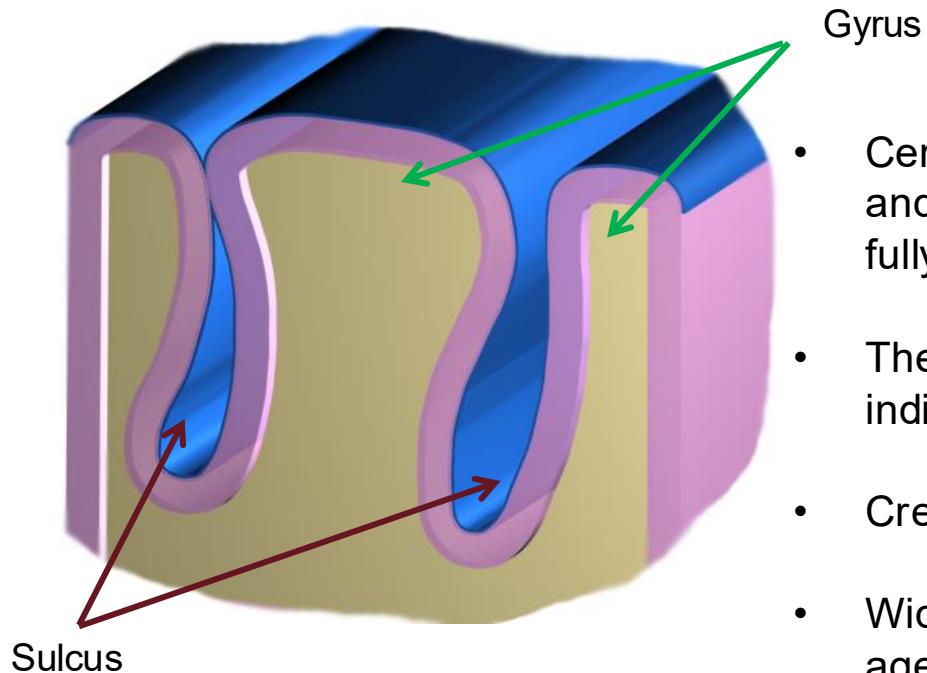




Sulci morphometry

Brain Sulci

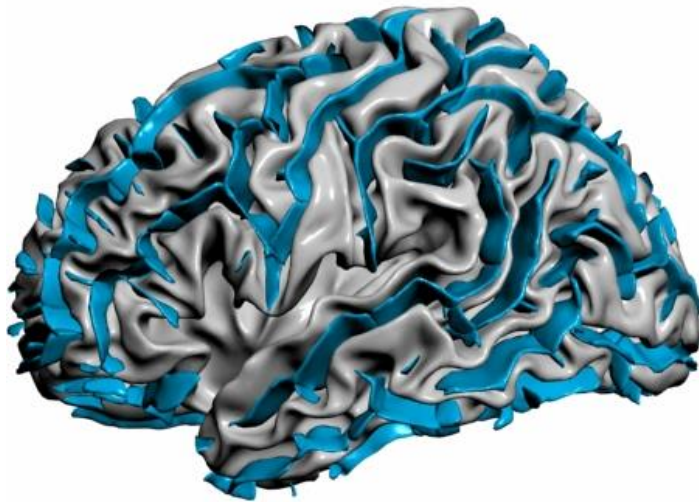
- A *sulcus* is a depression or groove in the cerebral cortex



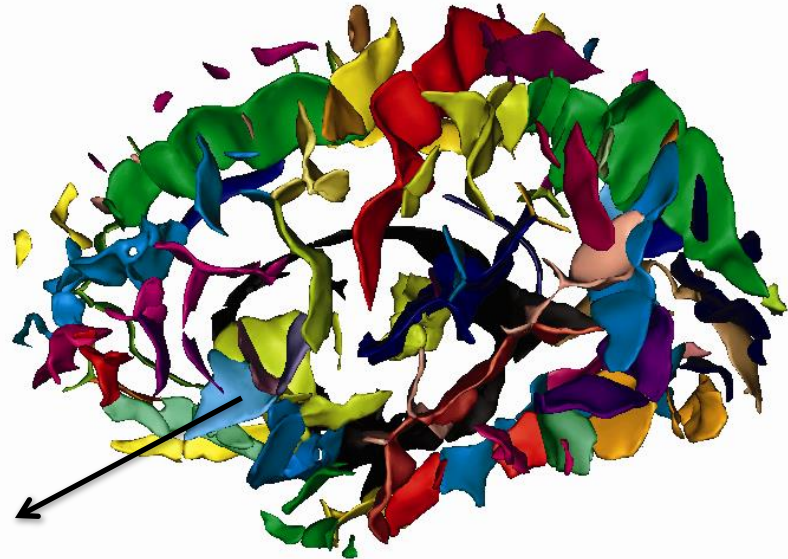
- Cerebral convolutions appear at about 5 months and take at least into the first year after birth to fully develop
- The sulcal pattern varies between human individuals
- Create a larger surface area
- Width of cortical sulci not only increases with age, but also with cognitive decline in the elderly

Brain Morphometry: Sulci extraction

Sulci Extraction



Sulci Labelling



Sulcal Measures

Sulcal Length



Sulcal Depth



Sulcal Width

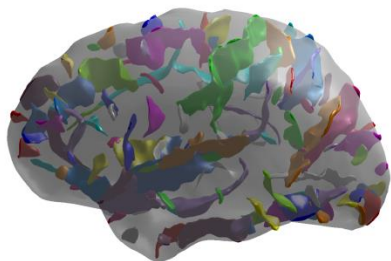


Sulci metrics: Brainvisa

Median sulcal surface



Median sulcal surface
and
Hull surface



Sulcal roots



Sulcal Length



Distance of the median sulcal surface intersecting the hull

Sulcal Depth



Geodesic distance (along sulcal roots) between the fundus and the hull

Sulcal Width



Fold opening

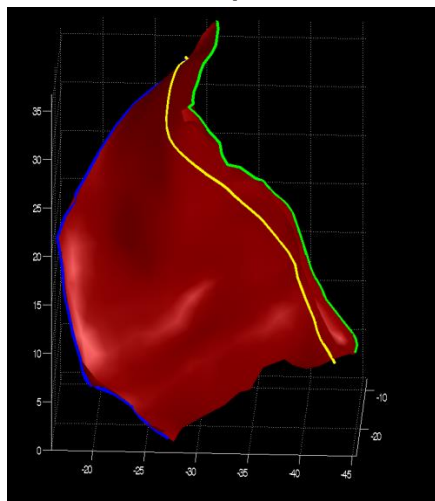
It is the volume of CSF in the sulcus, divided by the surface of the skeleton mesh

Sulci Metrics: Sulcal Length

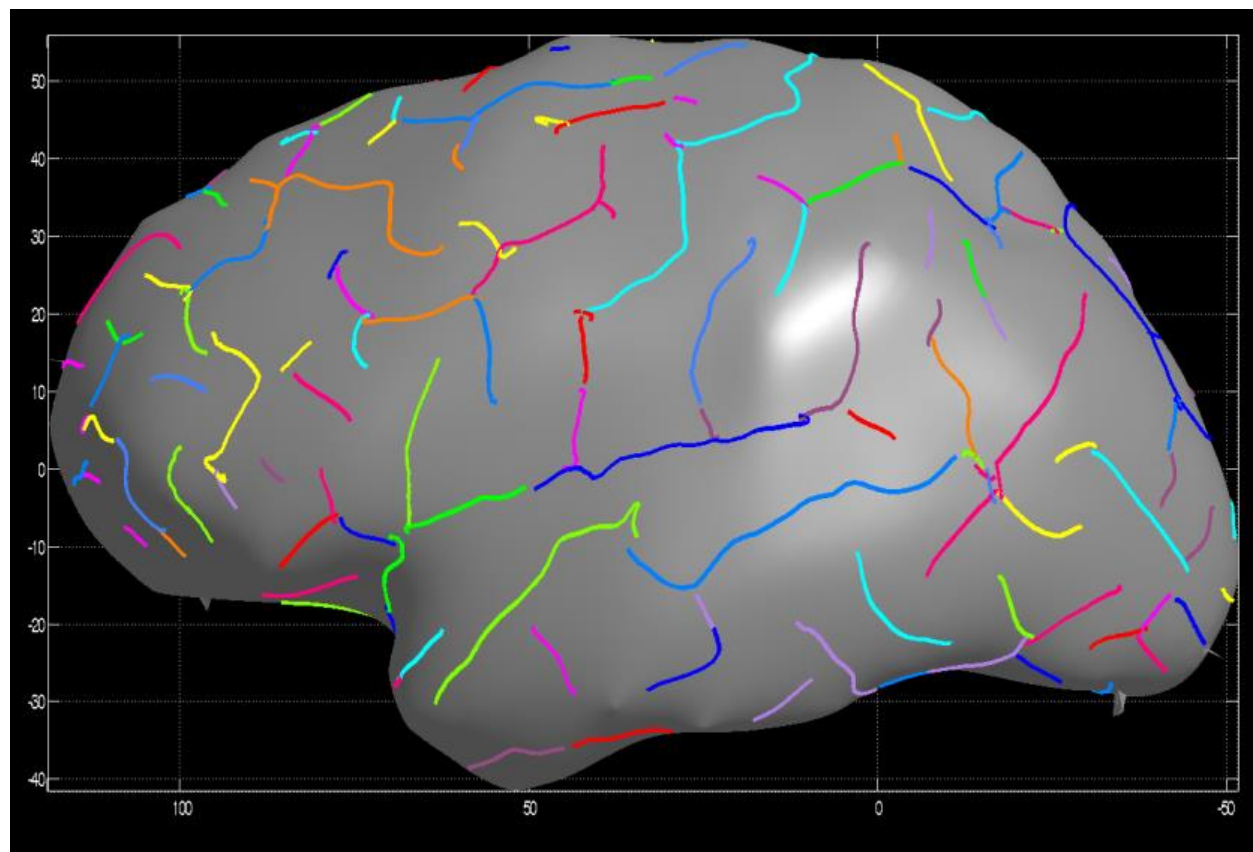
Sulcal Length



- Top line
- Bottom line
- Interception line



Hull and Interception Lines

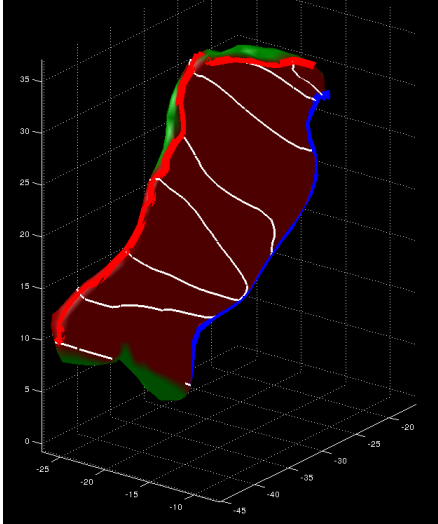


Sulci Metrics: Sulcal Depth

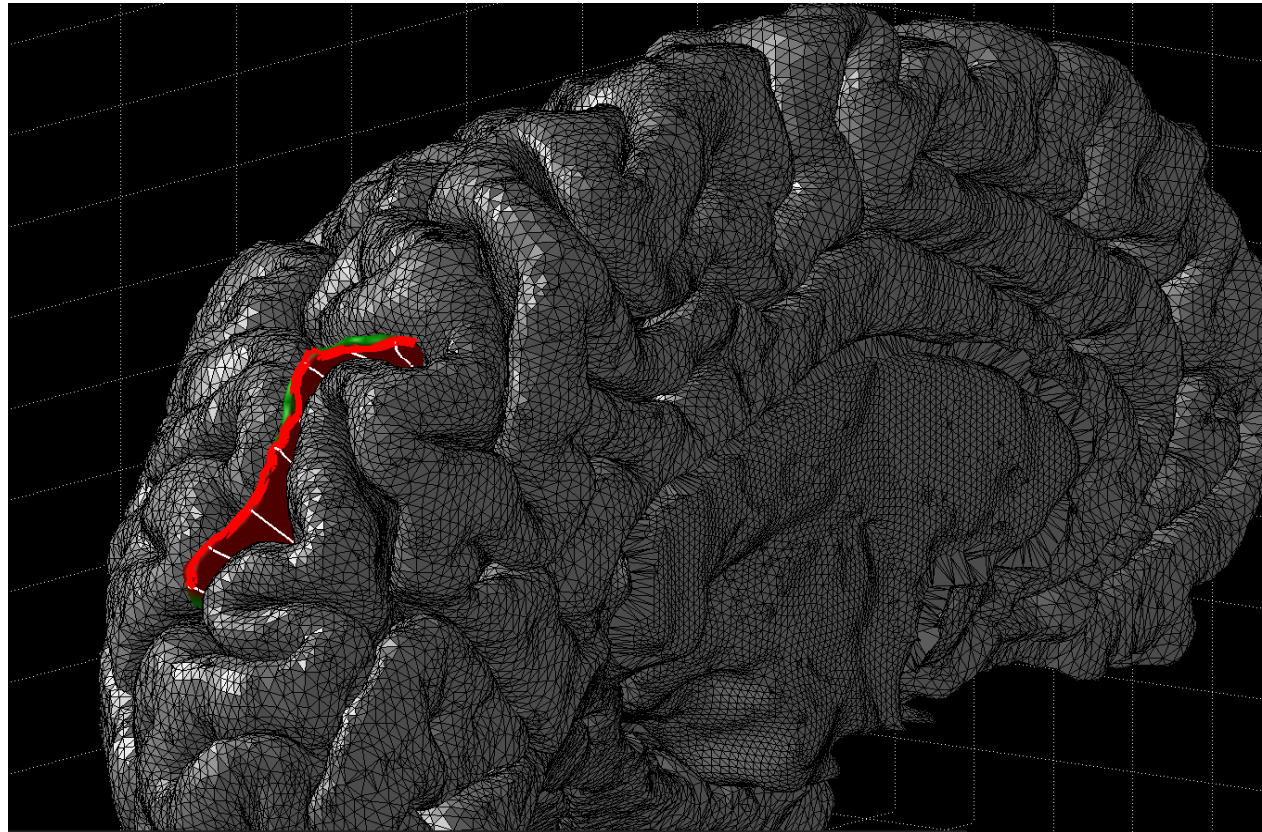
Sulcal Depth



— Top line
— Bottom line



Pial, median mesh and depth lines

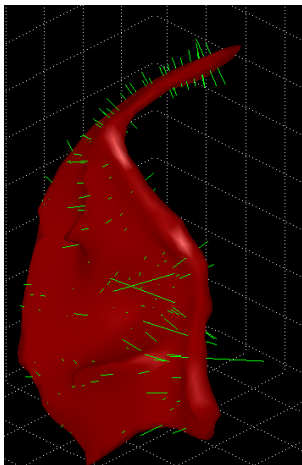


Sulci Metrics: Sulcal Width

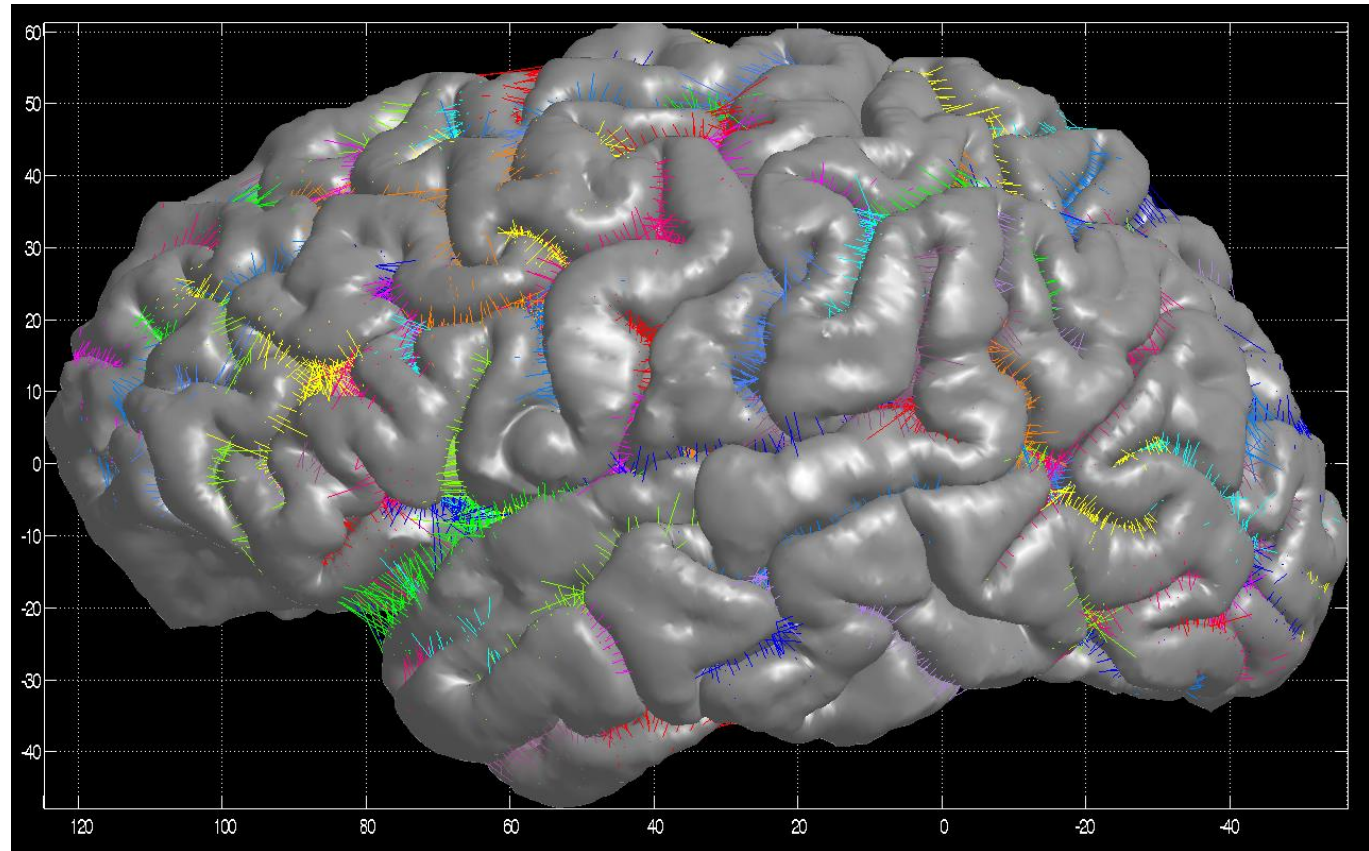
Sulcal Width

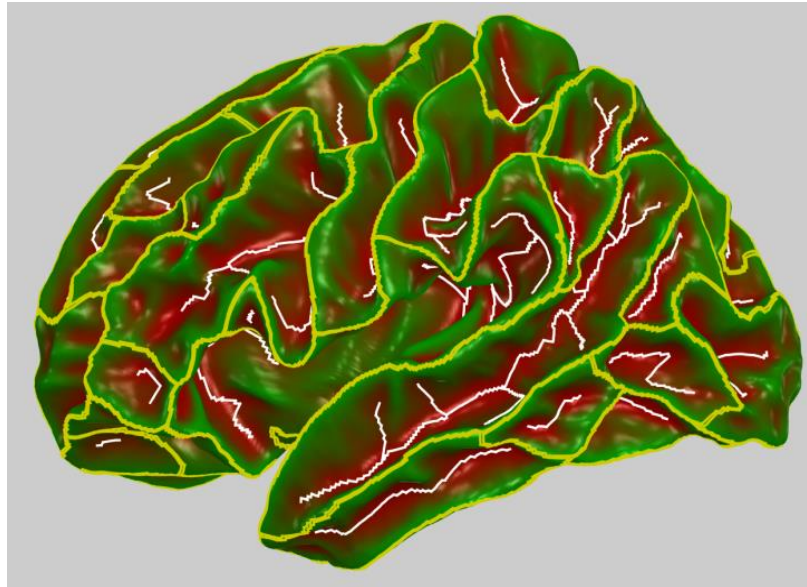


 Width lines



Pial and sulcal width lines

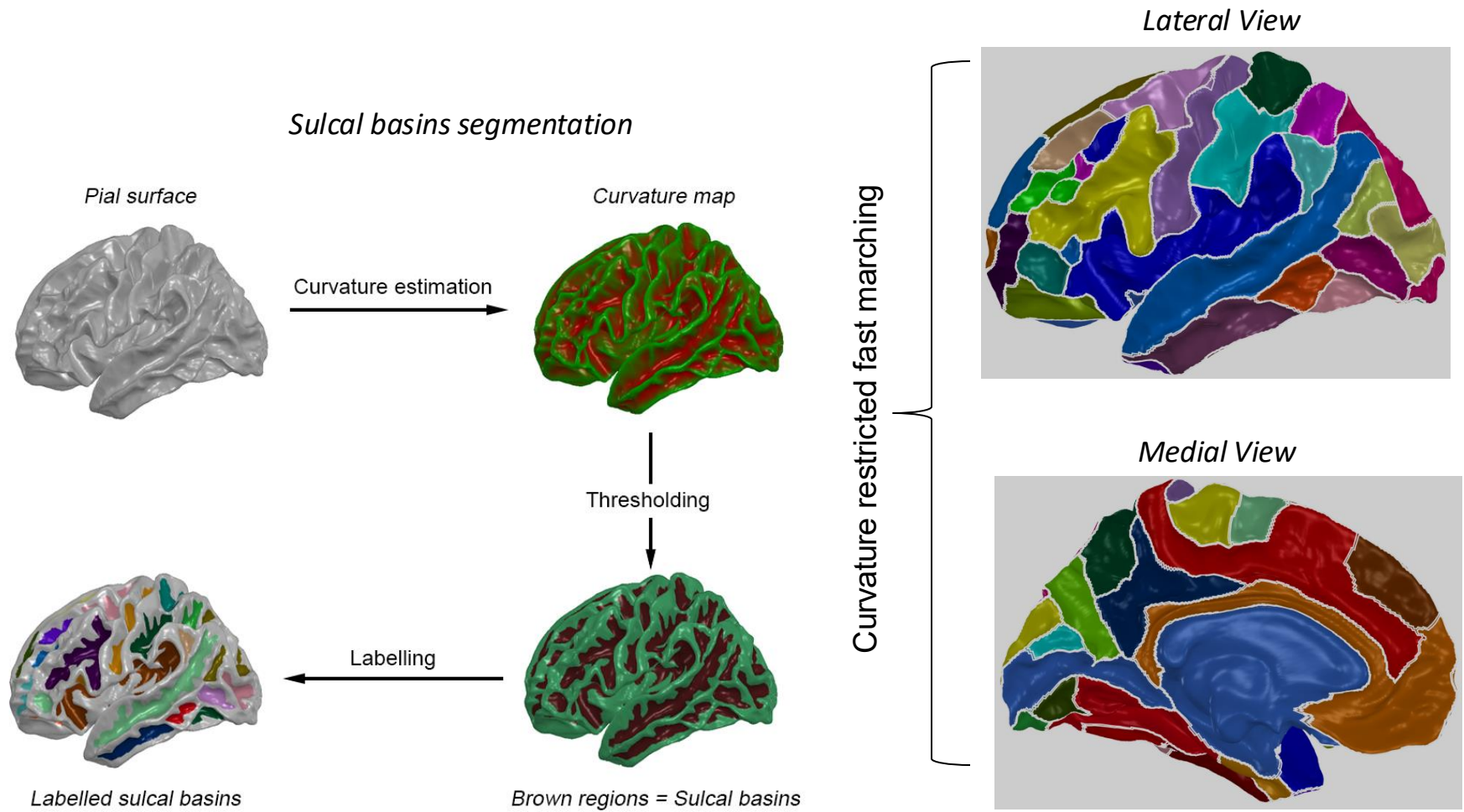




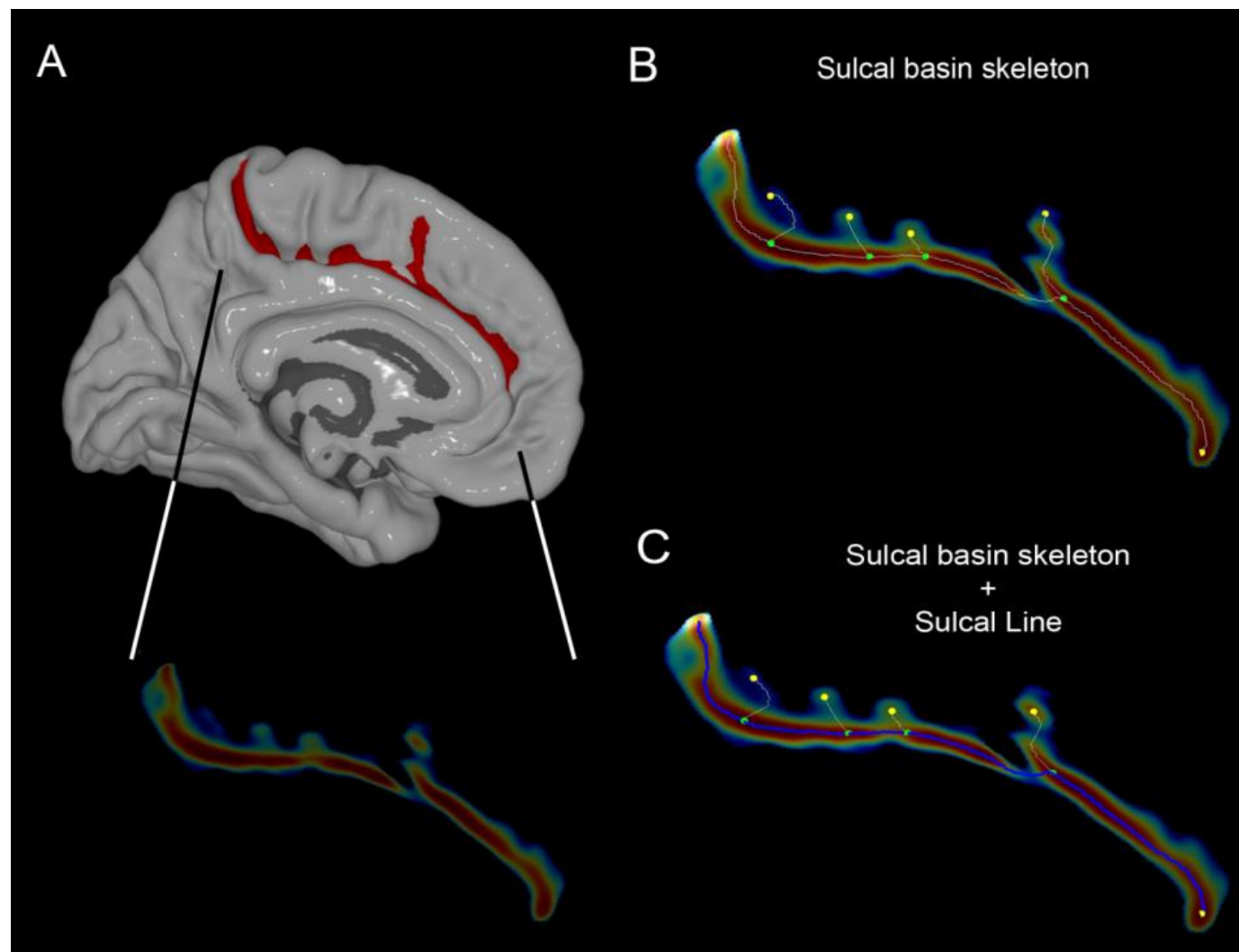
Sulcal and gyral crown lines

Fernández-Pena, A., de Blas, D., Navas-Sánchez, F. J., Marcos-Vidal, L., Gordaliza, P. M., Santonja, J., Janssen, J., Carmona, S., Desco, M., & **Alemán-Gómez, Y.** ABLE: Automated Brain Lines Extraction Based on Laplacian Surface Collapse. **Neuroinformatics**. 2022; 1–18

Gyral Crowns

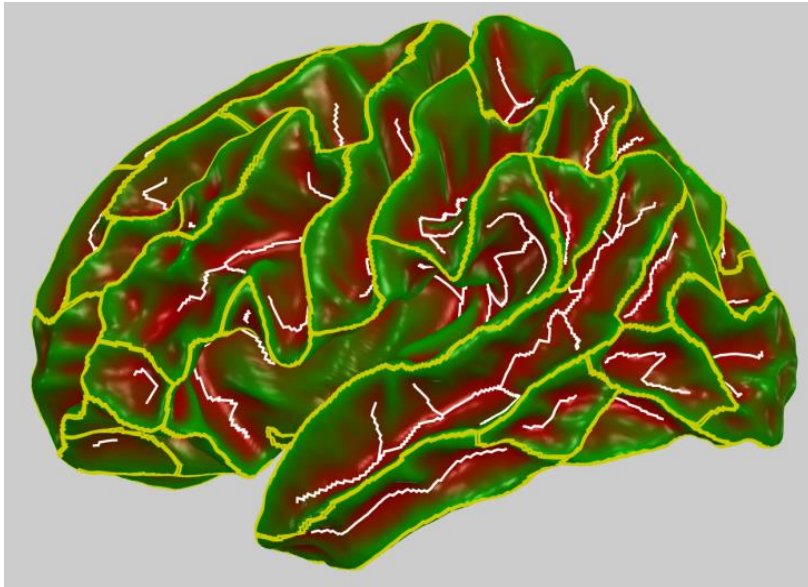


Sulcal lines extraction

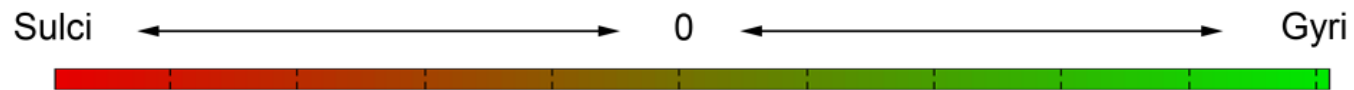
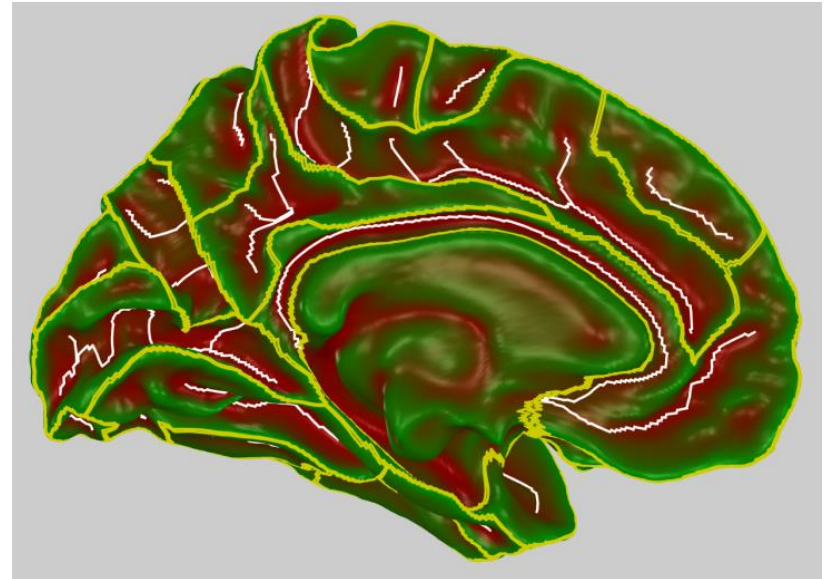


Sulcal lines extraction

Lateral View

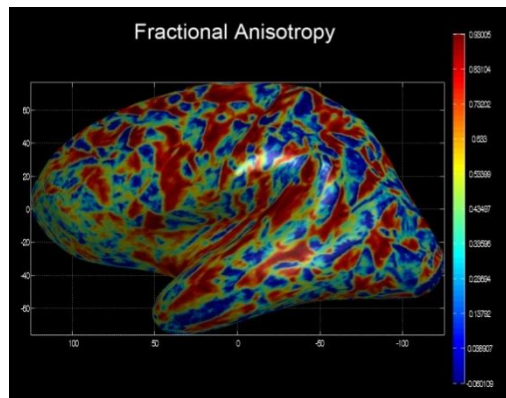


Medial View

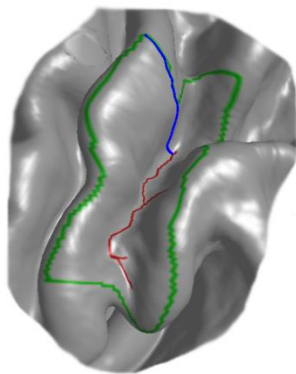


Morphometric Profile Construction

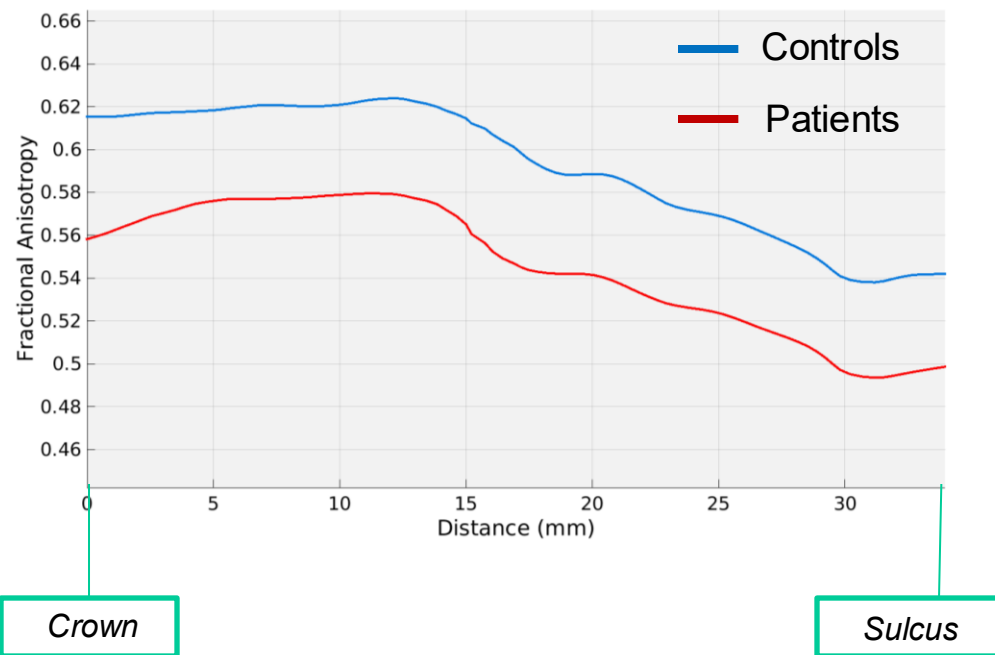
Surface FA



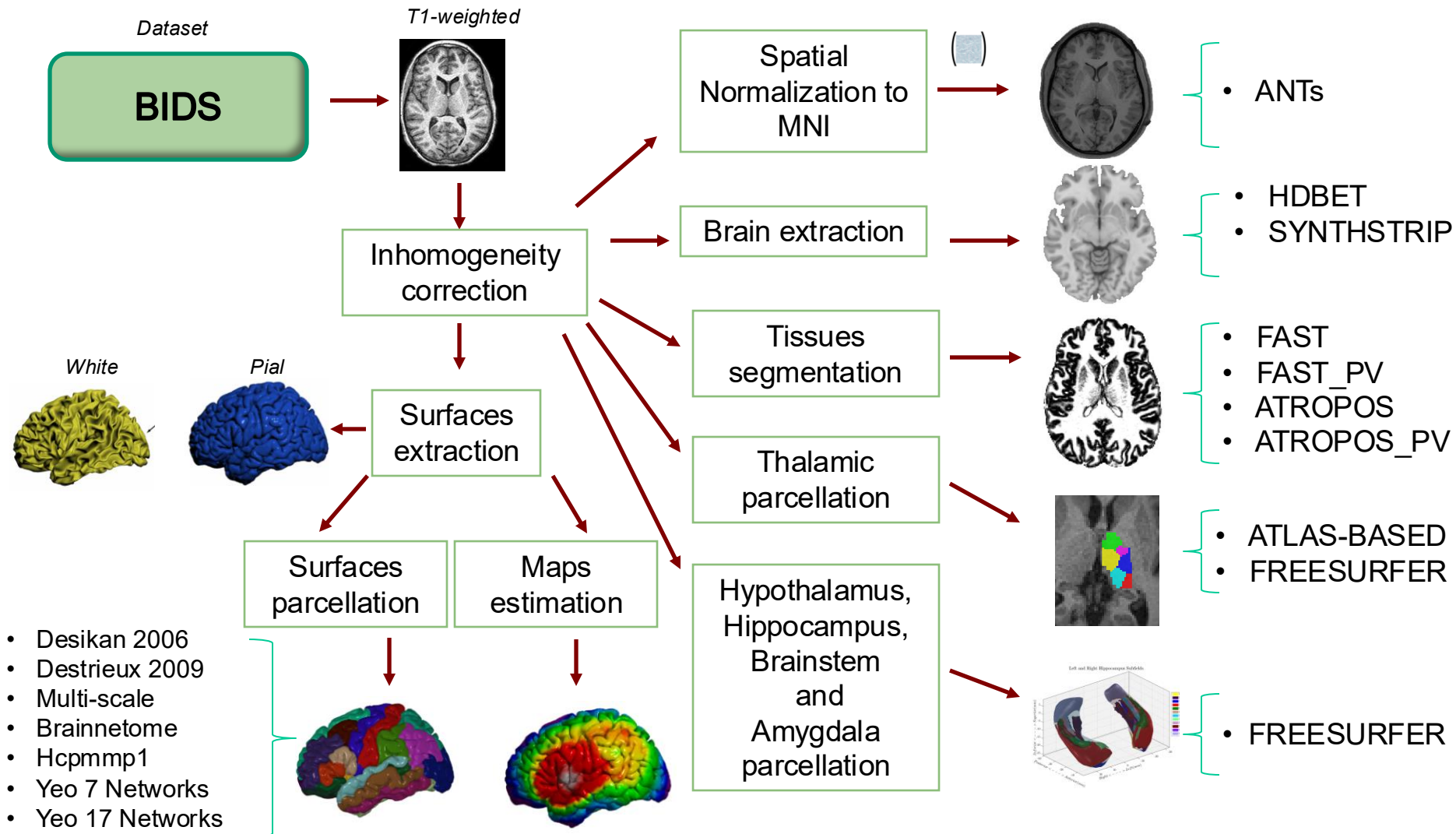
*Path between
Sulcal Line and Gyral Crown*



Fractional anisotropy profile



Processing for T1-weighted images



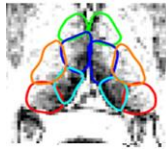
Derivatives: Metrics

Regional metrics

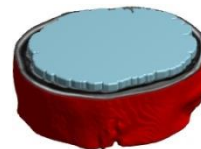
Surface Area ^[1]



Mean GM concentration



Intracranial volume

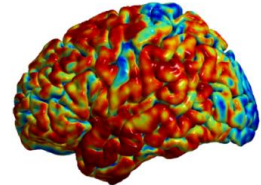


Vertex/voxel-wise metrics

GM Concentration^[5]



Cortical thickness ^[6]



Gyrification Index ^[2,3]

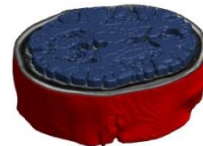
Pial Surface

Hull Surface

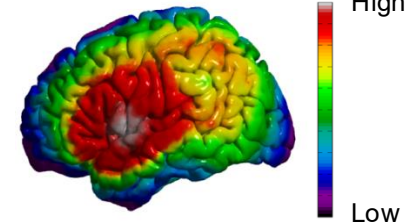


$$GI = \frac{\text{Pial Surface Area}}{\text{Hull Surface Area}} > 1$$

Brain volume

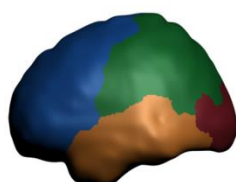
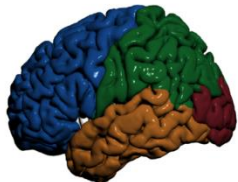


Local Gyrification Index ^[7]



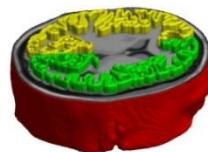
Pial Surface

Hull Surface

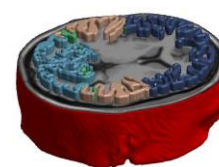


$$rGI = \frac{\text{Pial Surface Area}_{(\text{Lobe})}}{\text{Hull Surface Area}_{(\text{Lobe})}} > 1$$

Tissues



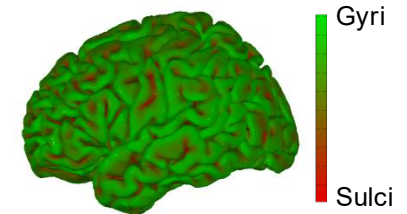
Lobes



Structures



Curvature Index ^[8]



[1] Desikan et al. 2006. Neuroimage

[2] Van Essen et al. 2001. JAMIA

[3] Su et al. 2013. Human Brain Mapping

[6] Roche et al. 2014. MICCAI

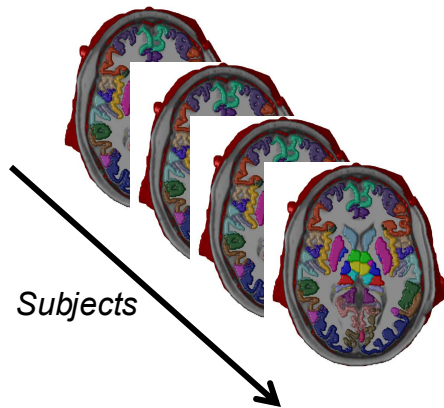
[6] Fischl et al. 2000. PNAS

[7] Schaer et al. 2008. IEEE Trans. Med. Imag.

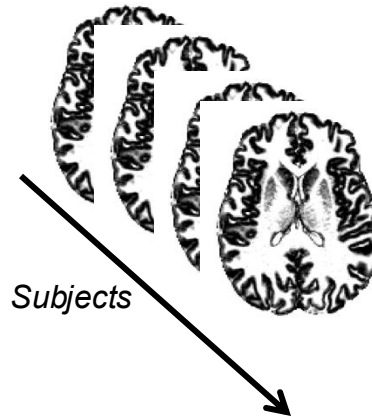
[8] Dale et al. 1999. Neuroimage

Brief summary

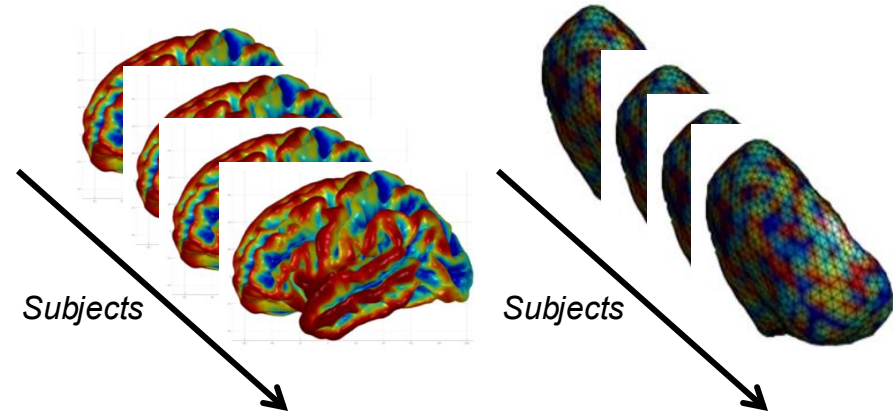
ROI analysis



Voxel-Based Morphometry

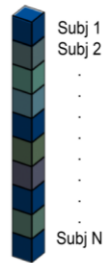


Surface-Based Morphometry



For each ROI, voxel or vertex

Group1

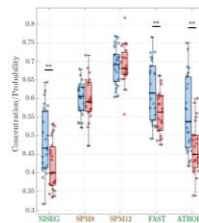


Group2

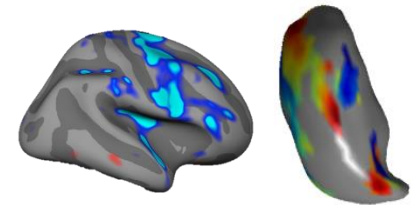
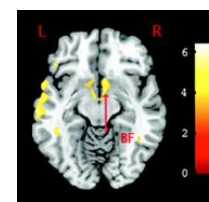


VS

Statistics

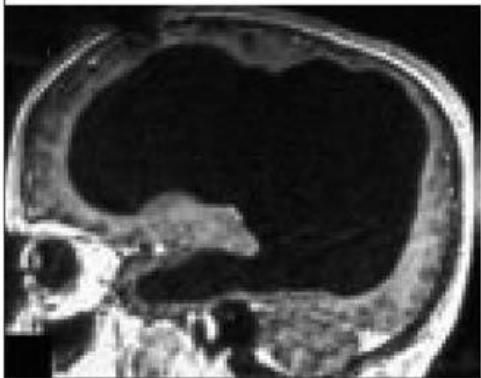
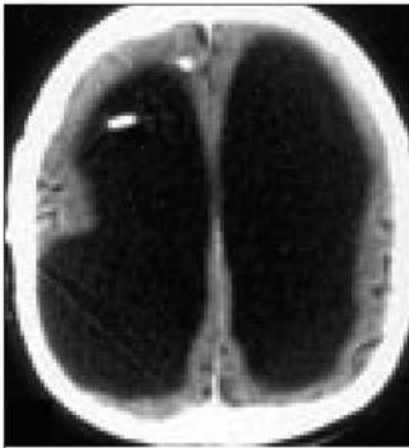


Results

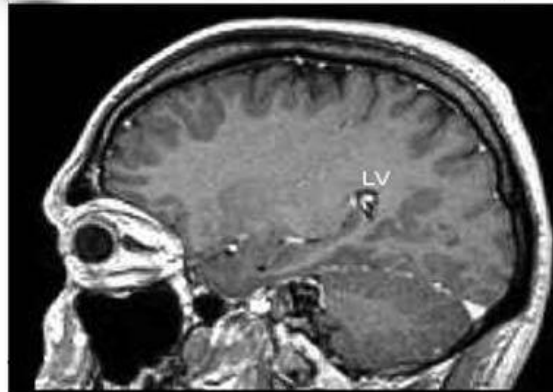
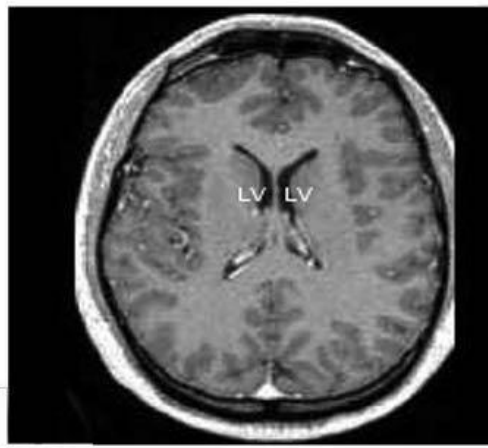


Interesting cases (I)

Sex: Male
Age: 44 yo
IQ: 75 (below the mean)



Healthy control



The Lancet

Interesting cases (II)

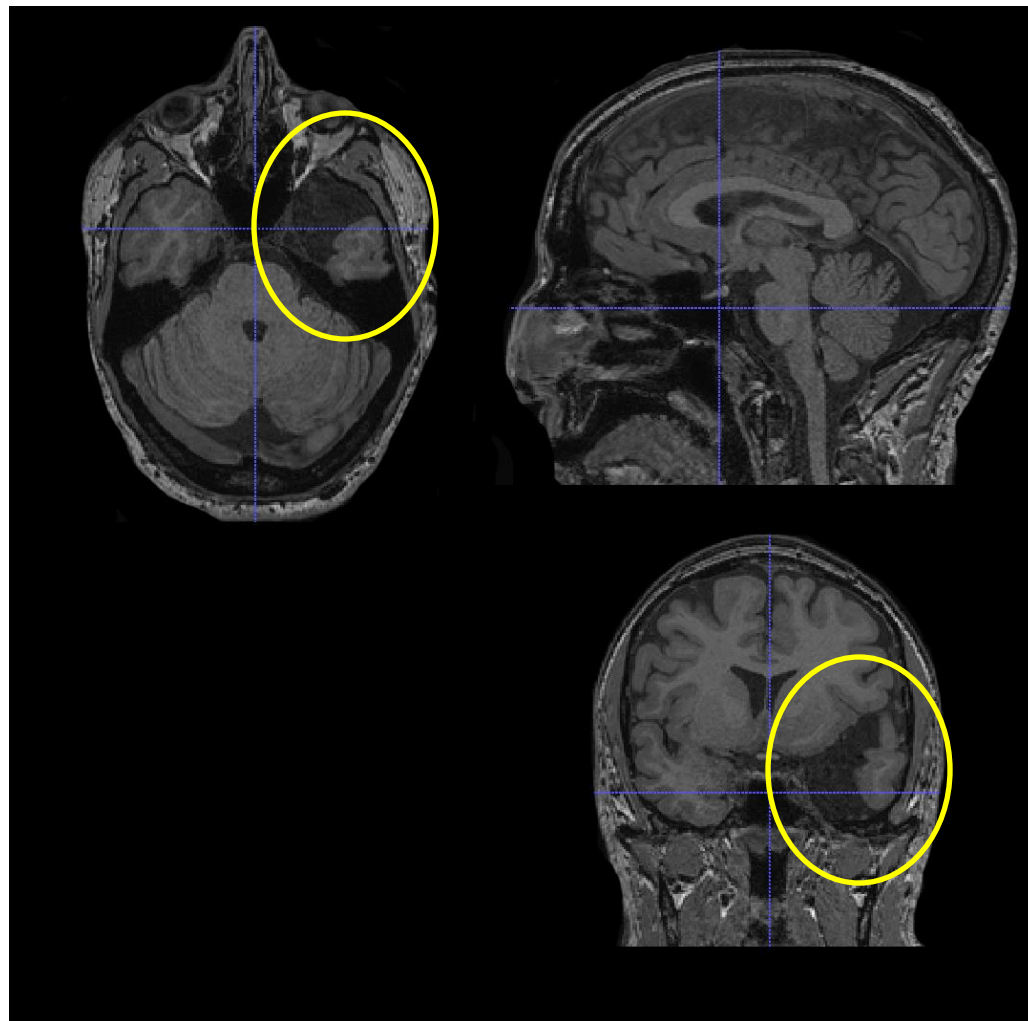
Sex: Female
Age: 24 oy



Healthy control

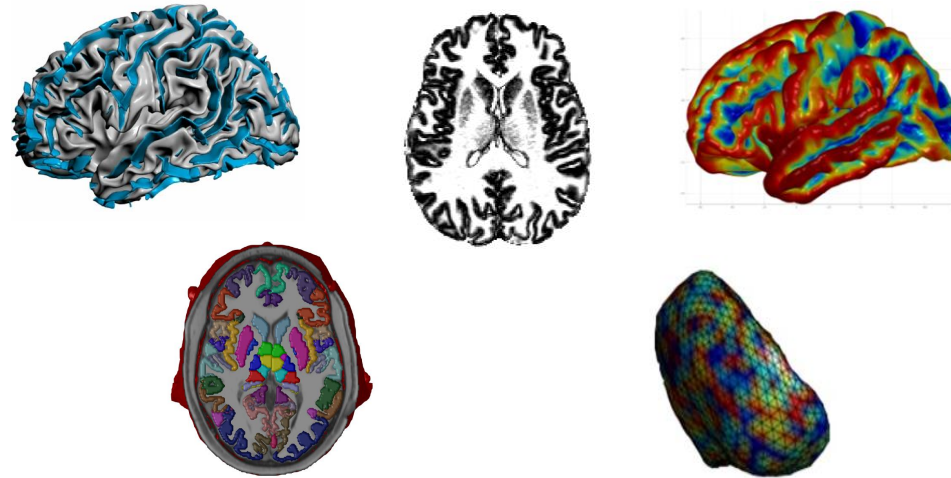
Interesting cases(III) HGGM

Sex: Male
Age: 24 yo



References

- Ashburner & Friston (2005) *Unified Segmentation*. NeuroImage 26:839-851
- Mechelli et al. (2005) *Voxel-based morphometry of the human brain...* Current Medical Imaging Reviews 1(2)
- Ashburner & Friston (2009) *Computing average shaped tissue probability templates*. NeuroImage 45(2):333-341
- Dale, A. M., B. Fischl, et al. (1999). "Cortical surface-based analysis. I. Segmentation and surface reconstruction." Neuroimage. 9(2): 179-194.
- Davatzikos, C. and N. Bryan (1996). "Using a deformable surface model to obtain a shape representation of the cortex." IEEE Transactions on Medical Imaging. 15(6): 785-795.
- Fischl, B., M. I. Sereno, et al. (1999). "Cortical surface-based analysis. II: Inflation, flattening, and a surface-based coordinate system." Neuroimage. 9(2): 195-207.
- Thomas E. Nichols: <http://www.sph.umich.edu/~nichols/FDR/>
- Jeanette Mumford: <https://www.youtube.com/channel/UCZ7gF0zm35FwrFpDND6DWeA>
-



MRI Processing Theory: Structural Morphometry

Yasser Alemán-Gómez

Radiology department. CHUV, Switzerland
October 1st, 2026